Net-Aware Learned Block Mapping for Reversible Data Hiding in Binary Images

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ABSTRACT

This paper presents a reversible data hiding scheme for plaintext binary images under an explicit synchronization model. Its technical contribution is serialization-aware net-capacity-driven activation of reversible PEAK/ZERO 3×3 mappings: the active mapping configuration is selected by usable mapping capacity after charging the exact serialized description required for deterministic recovery. The implementation represents the active PEAK subset by a combinatorial rank and the distance-one flip positions by base-9 digits; a convolutional neural network (CNN) supplies the learned distortion ranking for admissible ZERO candidates. On eight 512×512 document images, this activation raises mean net payload from 2139 to 2245 bits relative to an otherwise identical wire-charged gross-selection variant and preserves exact round-trip reversibility. Frozen external validation on all 199 scanned FUNSD documents, with no training or tuning on FUNSD, preserves exact message and cover recovery for every available run under both fixed-threshold and Otsu binarization. Relative to all-feasible wire-charged activation, the serialization-aware variant saves about 239–241 auxiliary bits per paired image and is never worse in delivered net payload, although positive net payload remains uncommon at targets up to 512 bits. In a common executable protocol on 100 disjoint thresholded BOSSbase images, the proposed adaptive block-mapping method (ABM), Dong, Huynh–Nguyen, and the opposite-center-pair method (PPOCP) are compared with identical messages, payload targets, distance-reciprocal distortion (DRD) computation, auxiliary-bit accounting, and reversibility checks. The proposed method achieves the highest useful capacity among the evaluated implementations, while PPOCP retains the lowest-distortion and most favorable detectability profile. Logistic detector AUC reaches 0.900 and 0.957 at 256 and 512 bits, placing the validated operating region in capacity-oriented reversible annotation over a lossless channel with detectability-aware payload selection.

1. Introduction

The proliferation of digital archives has increased the need to associate authentication data, annotations, and integrity evidence with images while preserving the original content. Cryptography protects message content [1], whereas steganography conceals its presence [2]. Reversible data hiding (RDH) combines these objectives with exact cover restoration, a property that is especially valuable for medical, legal, and administrative binary documents.

Binary images have two pixel states, so every modification can alter stroke continuity, topology, or local readability. Specialized RDH methods therefore rely on pattern statistics, overlapping neighborhoods, or reversible block substitutions rather than direct grayscale-domain operations [3–13]. Their practical performance depends on three coupled quantities: visual distortion, the side information required for exact synchronization, and the statistical footprint visible to a steganalyst.

Existing methods occupy complementary parts of the capacity–distortion space. Magnification-based embedding provides low distortion by increasing image size [8]; PPOCP uses overlapping opposite-center pairs for perceptually selective changes [9]; Huynh–Nguyen exploit the combinatorial capacity of 3×3 block mapping [4]; and Dong et al. adapt overlapping patterns to image statistics [7]. These contributions motivate a unified evaluation in which useful payload is measured after serialized metadata and all methods use the same messages, cover images, and distortion metric.

This work builds on established net-capacity accounting. The distinguishing algorithmic decision is that the exact serialized description cost of the reversible mapping configuration participates in deciding which PEAK/ZERO tables remain active. A convolutional neural network ranks distance-one ZERO candidates by predicted perceptual cost, while the reversible mechanism remains a deterministic PEAK/ZERO substitution with an enumerative auxiliary stream. Gaussian-process threshold exploration and the Random Forest uniform-block agent are reported as diagnostic ablations; the deployed distance-one codec consists of the deterministic PEAK/ZERO mapping, CNN ranking,

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serialization-aware activation, and enumerative auxiliary stream.

The main scientific contributions of this paper are:

1. Serialization-aware net-capacity-driven activation of reversible PEAK/ZERO mappings, where active tables are selected by usable mapping capacity after the exact serialized description cost required for deterministic recovery.
2. An executable enumerative wire representation for the active PEAK subset and the associated distance-one flip positions, yielding a decoder contract $(I', A_{\mathrm{wire}}) \mapsto (I, M)$.
3. A learned candidate-cost ranker evaluated against Hamming-index, direct DRD-cost, transition-preserving, and connectivity-aware rankers, plus a CNN architecture ablation.
4. A common executable protocol comparing the proposed adaptive block-mapping method (ABM), the opposite-center-pair method (PPOCP), Huynh–Nguyen, and Dong with identical covers, messages, payload targets, reversibility checks, distance-reciprocal distortion (DRD) implementation, auxiliary-bit accounting, paired statistics, resources, and grouped steganalysis.
5. Frozen external validation of the deployed CNN-ranked codec on all 199 real scanned FUNSD documents under fixed-128 and Otsu binarization, without retraining or tuning, including exact round-trip, availability, auxiliary-cost, net-payload, DRD, and paired A1/A2 activation analyses.

Section 2 positions the work in binary RDH and steganalysis. Section 3 defines the reversible mapping, learned ranking, and net-capacity codec. Section 4 describes the common protocol and reports document-image, BOSSbase, and FUNSD results. Section 5 synthesizes the practical operating region, validity conditions, and remaining limitations, and Section 6 concludes the study.

2. Related Work

Binary-image RDH is constrained by the lack of grayscale redundancy: each pixel flip can break strokes, isolate components, or change local topology. The relevant literature therefore centers on pattern substitution, distortion control, and the side information required to restore the cover exactly.

2.1. Binary Pattern and Block Substitution

Early binary-image RDH uses logical operations and run-length or pattern statistics [13]. Ho et al. [14] established high-capacity reversible pattern substitution by exchanging frequent and eligible replacement patterns. Dong et al. [7] refine this line through adaptive overlapping patterns, PF/PFR handling, and a compressed location map. Yin et al. [9] select opposite-center pattern pairs (PPOCP), yielding a low-distortion profile. Zhang et al. [8] obtain very low distortion through image magnification, but the carrier dimensions change. Huynh and Nguyen [4] are the closest direct block-mapping reference: they use non-overlapping 3×3 blocks, frequent PEAK patterns, zero-frequency replacement patterns, and Hamming-distance-limited mappings. These works establish pattern substitution, 3×3 PEAK/ZERO block mapping, and Hamming-distance control as the relevant technical foundation.

Other binary-image variants expand the design space. Chhajed and Garg [3] use block-diagonal partition patterns, Ren et al. [6] use sliding windows, and Yang [5] prioritizes mixed black-white regions. They are useful contextual comparators, with executable wire models left open for a direct net-payload benchmark under the protocol used here.

2.2. Distortion, Coding, and Reconstruction Information

Binary document quality is measured with DRD because it accounts for spatially non-uniform perceptual sensitivity [15]. Feng, Lu, and Sun [12] further show that run continuity and connectivity are important when choosing flippable pixels. These results motivate the ranker ablations in Section 4.3: Hamming ordering alone is compared with direct DRD-cost, transition-preserving, connectivity-aware, and CNN-based rankings.

Coding-based RDH for binary covers also predates this work. Zhang et al. [16] improve binary-cover RDH schemes using optimal codes, and Zhang et al. [17] study optimal transition probabilities under general distortion metrics. Xiao et al. [18] propose a general-distortion RDH framework for binary covers using reconstruction information and matrix embedding; Li et al. [19] extend this line with multi-embedding. These studies establish distortion-aware binary-cover coding and reconstruction information as important conceptual prior art. Their contribution is complementary to the plaintext 3×3 PEAK/ZERO block-mapping codec developed here, where the serialized mapping description itself determines which reversible mappings remain active.

2.3. Encrypted, Halftone, and Learned Neighbors

Encrypted-domain binary RDH uses a different carrier model, but it is relevant to the treatment of side information. Ren, Lu, and Chen [11] use prediction in encrypted binary images, and recent encrypted-domain work includes adaptive block coding for grayscale carriers [20]. Halftone RDH also records dynamic embedding-state information [21]. These neighboring methods confirm that side information and compression are established tools; the distinction in this paper is their use as an exact serialized cost inside the active mapping-selection rule.

Learning serves here as candidate-cost ranking, while the reversible encoder remains deterministic and table based. Learned RDH and steganalysis show why such ranking and detection analyses are relevant [22–25]. Because PEAK/ZERO substitution changes local pattern histograms and transition rates, the experiments include grouped detectors to delimit the payload region where marked images become statistically separable from covers.

Table 1
Comparative overview of binary RDH and related systems

Method (Year)	EC (bits)	Net accounting	PSNR (dB)	DRD	Adaptive	Uniform blocks	Steg. eval.
Ho et al. 2009 [14]	Published	Open	–	–	Yes	Pattern-dependent	Not extracted
Zhang et al. 2012 [16]	Code-dependent	Reconstruction info	–	Distortion bound	Yes	Binary cover sequence	Not extracted
Yin et al. 2020 [9]	1157	Open	27.17	0.23	Fixed	Reserved	Not extracted
Dong et al. 2020 [7]	~1500	Open	~25	<0.5	Yes	Reserved	Not extracted
Xiao et al. 2022 [18]	Variable	Reconstruction info	High	Optimized	Yes	Pixel sequence	Not extracted
Li et al. 2023 [19]	1000 example	Reconstruction info	49.45	–	Yes	Pixel sequence	Not extracted
Ren et al. 2023 [6]	~2000	Open	~24	<1	Fixed	Reserved	Not extracted
Zhang et al. 2019 [8]	369	Open	32.31	0.08	Fixed	Reserved	Not extracted
Chhajed & Garg 2023 [3]	<2000	Open	~25	<1	Fixed	Reserved	Not extracted
Yang 2023 [5]	>2000	Open	~23	<1	Partial	Reserved	Not extracted
Huynh & Nguyen 2025 [4]	2070	Open	24.13	0.55	Fixed	Reserved	Not extracted
Proposed	**3832 gross cap.**	**2245 net at** $q = 2874$	–	**0.76**	**Yes**	**Evaluated**	**Yes**

EC: capacity reported by the cited study on its stated 512×512 protocol. Values provide cross-protocol context because datasets, payload definitions, and side-information accounting differ. "Steg. eval.": whether a steganalysis result was extracted for this overview table; Section 4.12 reports the common-protocol detector evaluation. For the proposed method, 3832 is the rounded mean A2 gross mapping capacity, $q = 2874$ is the mean embedded application payload at the 75% budget, and 2245 is the resulting mean delivered net payload. DRD is the reported common-protocol distortion summary; per-image PSNR and changed-pixel outputs are included in the reproducibility artifacts.

2.4. Research Gap

Table 1 consolidates the direct and conceptual neighbors. The relevant gap is an executable plaintext binary-image PEAK/ZERO method in which the active reversible mapping configuration is selected by usable mapping capacity after the exact serialized description cost needed for deterministic recovery. Within the reviewed corpus, this study contributes the joint treatment of serialization-aware activation, an enumerative PEAK/flip-position wire representation, and a common executable comparison against representative binary-image baselines.

3. Net-Aware Learned Distance-One Block Mapping

3.1. Overview and Design Rationale

This section specifies the complete system evaluated in Section 4. The design uses the established 3×3 PEAK/ZERO block-mapping family, but the active mapping tables are chosen by usable mapping capacity after the exact serialized mapping description is charged. The encoder and decoder are therefore defined as the pair of maps

$$\mathrm{Enc}(I, M) \to (I', A_{\mathrm{wire}}) \tag{1}$$

and

$$\mathrm{Dec}(I', A_{\mathrm{wire}}) \to (I, M). \tag{2}$$

The auxiliary stream A_{wire} is stored or transmitted alongside the marked binary image I' as explicit synchronization metadata. The unconstrained adaptive threshold is retained as an ablation that characterizes the broader capacity–distortion landscape; the deployed codec uses globally disjoint Hamming-distance-one substitutions. Consequently, each active block carries one bit and changes at most one pixel.

The evaluated system proceeds in four deterministic steps:

- learns a perceptual ordering of absent distance-one patterns;
- enforces a globally injective PEAK/ZERO assignment;
- selects the frequency-ordered table prefix that maximizes map-level net capacity; and
- serializes the payload length, active PEAK subset, and flip positions with an enumerative codec.

The embedding and extraction workflows are illustrated in Figures 1 and 2 and formalized in Algorithms 1 and 2.

3.2. Reversible Mapping Construction

Block representation and pattern histogram

Let I be a binary image of size $M \times N$. Complete non-overlapping 3×3 blocks are visited in raster order:

$$B_k = \{b_{i,j}\}, \quad i, j \in \{0, 1, 2\}, \quad k = 1, \dots, \left\lfloor \frac{M}{3} \right\rfloor \left\lfloor \frac{N}{3} \right\rfloor. \tag{3}$$

The resulting block count is the product of the two complete-block counts:

$$B = \left\lfloor \frac{M}{3} \right\rfloor \left\lfloor \frac{N}{3} \right\rfloor. \tag{4}$$

Each block is mapped to a decimal index:

$$d_k = \sum_{i=0}^{8} b_i 2^{8-i}, \tag{5}$$

where $b_i \in \{0, 1\}$ follows a fixed row-major ordering shared by both embedding and extraction; the first row-major bit is the most significant bit. Rows with indices at least $3\lfloor M/3 \rfloor$ and columns with indices at least $3\lfloor N/3 \rfloor$ are copied unchanged and never participate in capacity or extraction.

Uniform blocks ($d_k = 0$ and $d_k = 511$) are reserved for the separately evaluated Random Forest layer in Section 3.5; the core block-mapping histogram uses $d \in \{1, \dots, 510\}$. A histogram $H(d)$ is constructed for $d \in \{1, \dots, 510\}$.

Pattern sets

The implementation considers every non-uniform pattern occurring at least twice as an initial PEAK candidate. Candidates are sorted by decreasing frequency and then by numerical index. The pattern space is divided into:

- **PEAK candidates** $\mathcal{P}_0 = \{d : H(d) \geq 2, d \notin \{0, 511\}\}$,
- **Unused patterns (ZEROs)** $\mathcal{Z}$: patterns such that $H(d) = 0$,
- **Residual patterns**: patterns outside the embedding tables.

The final active set $\mathcal{P} \subseteq \mathcal{P}_0$ is selected by the explicit map-level net-capacity rule defined below.

Pattern distance analysis

For each frequent pattern $P_i \in \mathcal{P}$ and each unused pattern $Z_j \in \mathcal{Z}$, the Hamming distance between their 3×3 representations is computed:

$$D_{i,j} = \sum_{p=1}^{9} \delta(P_i[p], Z_j[p]), \tag{6}$$

where $\delta(a, b) = 1$ if $a \neq b$ and 0 otherwise.

The set $\{D_{i,j}\}$ characterizes the distortion cost associated with mapping P_i to available unused patterns.

Adaptive candidate selection

Candidate patterns for each frequent pattern are selected adaptively. Let μ_i and σ_i denote the empirical mean and standard deviation of $\{D_{i,j}\}$. The adaptive Hamming threshold and candidate set are

$$H_{T,i} = \mu_i + \alpha\sigma_i \tag{7}$$

and

$$\mathcal{Z}_i = \{Z_j \in \mathcal{Z} \mid D_{i,j} \leq H_{T,i}\}, \tag{8}$$

where α controls the capacity–distortion trade-off.

This formulation uses empirical distance statistics rather than a parametric model for the observed distance values.

3.2.1. Globally Injective Mapping Construction

For the unconstrained ablation, let $k_i = |\mathcal{Z}_i|$ and define

$$L_i = \lfloor \log_2(k_i + 1) \rfloor \tag{9}$$

bits per occurrence. Exactly $2^{L_i} - 1$ candidates are retained:

$$\widetilde{\mathcal{Z}}_i \subseteq \mathcal{Z}_i, \qquad |\widetilde{\mathcal{Z}}_i| = 2^{L_i} - 1, \tag{10}$$

which makes the mapping

$$T_i : \{0, 1\}^{L_i} \longleftrightarrow \{P_i\} \cup \widetilde{\mathcal{Z}}_i \tag{11}$$

a genuine bijection. After a ZERO is assigned, it is removed from the available pool; therefore the ranges of all T_i are globally disjoint.

The deployed distance-one policy restricts $L_i = 1$. Symbol 0 preserves P_i, while symbol 1 uses the lowest predicted-cost unassigned ZERO satisfying $D(P_i, Z) = 1$. Thus every modified block flips exactly one pixel.

Deterministic block traversal order

To ensure reproducibility and unambiguous extraction, a deterministic block traversal order is enforced.

The image is traversed in raster-scan order (left-to-right, top-to-bottom) at the block level. Each block is visited exactly once during both embedding and extraction.

During embedding, frequent patterns are processed in descending order of occurrence. However, the spatial traversal order of blocks remains unchanged. Each block is modified at most once, and block decisions are spatially local once the mapping tables are fixed.

Multi-pattern embedding strategy

Embedding scans spatial blocks once in raster order. When a block pattern is an active PEAK, the next bit selects either the unchanged PEAK or its assigned ZERO. The last symbol is truncated to the declared payload length during extraction; for the deployed one-bit codec, the payload boundary is unambiguous.

The total embedding capacity is:

$$C_{\text{gross}} = \sum_{P_i \in \mathcal{P}} n_i L_i, \tag{12}$$

where n_i denotes the number of occurrences of P_i.

3.3. Embedding Procedure

The embedding procedure operationalizes the mapping construction above as a deterministic encoder that constructs a single marked image. It first freezes the spatial support: complete non-overlapping 3×3 blocks are processed in raster order, while border pixels are copied unchanged. This convention is used for two reasons. First, it gives the decoder the same block sequence with coordinate-free synchronization. Second, it prevents boundary handling from becoming hidden auxiliary information that would distort the net payload comparison.

The next stage builds the reversible alphabet available in the current cover. Patterns with at least two occurrences are eligible PEAKs, absent patterns are ZEROs, and all-white/all-black uniform patterns are kept in the reserved uniform-block pool because changing them tends to create isolated specks or holes in document backgrounds and strokes. For each PEAK, admissible replacements are restricted to distance-one ZEROs. This restriction makes every embedded symbol a single-pixel block modification, keeps the restoration rule local, and allows the auxiliary stream to store only one base-9 flip position per active table. At this point, the CNN ranks admissible ZEROs by predicted perceptual cost; a global disjointness check then prevents two PEAKs from sharing the same stego pattern.

The final selection stage is the serialization-aware part of the method. After candidate assignments have been fixed,

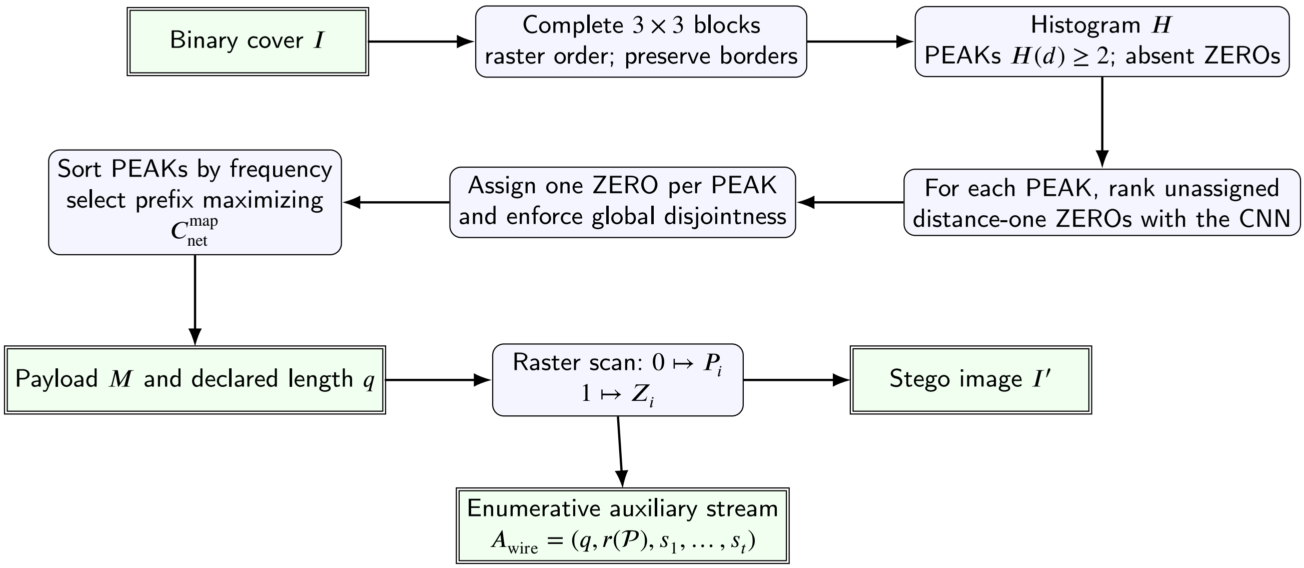

Figure 1: Evaluated embedding pipeline. Learned distance-one assignment is followed by explicit net-capacity optimization and enumerative serialization within a reversible-data-hiding synchronization model.

Algorithm 1 Net-Aware CNN-Ranked Distance-One Embedding

Require: Binary image I, payload $M = (m_1, \dots, m_q)$, trained ranker f_θ
Ensure: Stego image I', serialized auxiliary stream A_{wire}
Enumerate complete 3×3 blocks and compute $H(d)$
$\mathcal{P}_0 \leftarrow \{d : H(d) \geq 2,\ d \notin \{0, 511\}\}$; sort by $(-H(d), d)$
$\mathcal{Z} \leftarrow \{d : H(d) = 0\}$; $\mathcal{U} \leftarrow \emptyset$; $\mathcal{T} \leftarrow \emptyset$
for each $P_i \in \mathcal{P}_0$ **do**
  $C_i \leftarrow \{Z \in \mathcal{Z} \setminus \mathcal{U} : D(P_i, Z) = 1\}$
  if $C_i \neq \emptyset$ **then**
    $Z_i \leftarrow \arg\min_{Z \in C_i} (f_\theta(I, P_i, Z), Z)$
    $\mathcal{T}[P_i] \leftarrow \{0 \mapsto P_i, 1 \mapsto Z_i\}$; $\mathcal{U} \leftarrow \mathcal{U} \cup \{Z_i\}$
  end if
end for
Select the frequency-ordered prefix $\mathcal{T}^\star$ maximizing $C_{\text{net}}^{\text{map}}(\mathcal{S}) = \sum_{P_i \in \mathcal{S}} H(P_i) - |A_{\text{wire}}(\mathcal{S})|_2$
if $q > \sum_{P_i \in \mathcal{T}^\star} H(P_i)$ **then**
  reject the requested payload
end if
$I' \leftarrow I$; $j \leftarrow 1$
for each complete block of I in raster order **do**
  if its pattern is $P_i \in \mathcal{T}^\star$ and $j \leq q$ **then**
    replace it by $\mathcal{T}^\star[P_i][m_j]$; $j \leftarrow j + 1$
  end if
end for
Serialize q, the active PEAK subset rank, and base-9 flip positions
return I', A_{wire}

the encoder considers frequency-ordered prefixes of active mappings and selects the prefix that maximizes usable mapping capacity after charging the exact serialized description of that prefix. The message bits are then embedded during one raster scan: a payload bit 0 leaves an active PEAK block unchanged, while a payload bit 1 replaces it by the assigned ZERO. The declared message length q is serialized with the active table description, giving the decoder an explicit stopping point.

3.4. Efficient Extraction and Restoration

Extraction is designed as the exact inverse of the embedding contract, using the marked image and the serialized auxiliary stream. The auxiliary stream is decoded first because it fixes three pieces of synchronization information: the payload length, the active PEAK subset, and the single flip

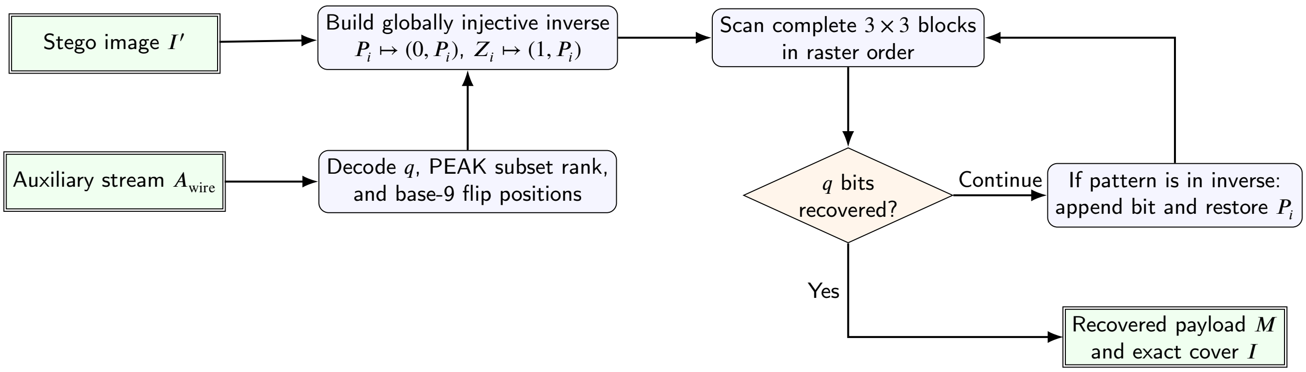

Figure 2: Extraction and restoration pipeline. The serialized payload length provides an unambiguous stopping condition, and incomplete border blocks remain untouched.

position that reconstructs each assigned ZERO. The decoder uses the serialized tables directly, making the transmitted synchronization contract the source of recovery decisions.

Once the active tables are reconstructed, they are collapsed into a single inverse dictionary. This is possible because the encoder enforces global injectivity: every stego pattern used by the codec belongs to at most one active mapping. Therefore, observing a PEAK pattern decodes bit 0 and restores the same PEAK, while observing its assigned ZERO decodes bit 1 and is also restored to the PEAK. Patterns outside the dictionary are copied unchanged. This dictionary view turns extraction into one linear raster scan with constant-time lookup.

The serialized payload length provides the stopping condition. After q bits have been recovered, remaining blocks are left unchanged because embedding used the same declared stopping point. If the scan finishes before q bits are recovered, the pair (I', A_{wire}) is rejected as inconsistent. This explicit rejection rule is necessary because exact reversibility is guaranteed for matched marked-image and auxiliary-stream pairs transmitted over a lossless channel.

To reduce extraction complexity, all mapping tables are merged into a single inverse dictionary:

$$\mathcal{D} : p \mapsto (m, P_i), \tag{13}$$

where p is a stego pattern, $m \in \{0, 1\}$ the extracted bit, and P_i the original pattern.

3.5. Learned Ranking, Ablations, and Wire Accounting

Distortion characteristics

For each embedded block mapped from P_i to $\mathcal{Z}_i$, the number of flipped pixels is bounded by:

$$\max_{Z_j \in \mathcal{Z}_i} D_{i,j}, \tag{14}$$

ensuring pattern-specific distortion control.

Gaussian-process diagnostic

The parameter α is explored by Gaussian-process Bayesian optimization. For every evaluated value, the implementation records capacity, PSNR, DRD, and the normalized objective

$$J(\alpha) = \lambda \left(1 - \frac{C_{\text{total}}(\alpha)}{C_{\text{ref}}}\right) + (1 - \lambda)\text{DRD}(\alpha). \tag{15}$$

Expected Improvement selects the next value of α [26]. The equal weighting $\lambda = 0.5$ gives the two normalized diagnostic terms identical influence; it serves only the exploratory scalarization of the unconstrained diagnostic. A λ sweep therefore characterizes the exploration stage, whereas all final tables use the independently specified distance-one policy. Optimizing α alone provides incomplete control of DRD $<$ 1, motivating the constrained distance-one policy used by the integrated system. The adaptive-threshold parameter α belongs to this unconstrained diagnostic; the deployed codec fixes the policy to Hamming distance one, and the final operating tables are governed by that fixed policy. The exploratory search uses $\alpha \in [0, 1.5]$, initial points $\{0, 0.5, 1, 1.5\}$, six EI iterations on a 151-point grid, $\lambda = 0.5$, exploration 0.01, and a Matérn-5/2 kernel with a white-noise term. The random seed is 20260607.

CNN-based candidate ranking

For a PEAK pattern and each distance-one ZERO candidate, the CNN receives three 9×9 channels: the centered PEAK, the centered candidate, and the average spatial context of all PEAK occurrences. It regresses the empirical reciprocal-distance replacement cost. The candidate with the smallest predicted cost is assigned to symbol 1, while symbol 0 preserves the PEAK.

The 9×9 support covers the central 3×3 block and a three-pixel margin in every direction. It contains the complete 5×5 DRD support around each of the nine possible flipped pixels, including pixels outside the embedding block; a 7×7 patch covers corner flips only partially. Larger patches add context beyond the local training target.

The network contains two 3×3 convolutional layers with 16 and 32 channels and ReLU activations, adaptive 3×3 average pooling, and a $288 \rightarrow 64 \rightarrow 1$ regressor with ReLU and Softplus. Training uses Smooth-L1 loss, AdamW with learning rate 10^{-3}, batch size 128, 20 epochs,

Algorithm 2 Deterministic Extraction and Exact Restoration

Require: Stego image I', serialized auxiliary stream A_{wire}
Ensure: Extracted message M, restored image I

Decode payload length q, active PEAKs, and their flip positions
Reconstruct all tables $T_i = \{0 \mapsto P_i, 1 \mapsto Z_i\}$
Build $\mathcal{D}[P_i] = (0, P_i)$ and $\mathcal{D}[Z_i] = (1, P_i)$
$I \leftarrow I'$; $M \leftarrow \emptyset$
for each complete block of I' in raster order **do**
    if $|M| = q$ **then**
        break
    end if
    Compute pattern index d
    if $d \in \mathcal{D}$ **then**
        Append the decoded bit to M and replace the block by its PEAK
    end if
end for
if $|M| \neq q$ **then**
    reject the inconsistent stego/auxiliary pair
end if
return M, I

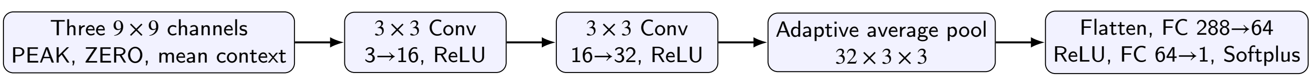

Figure 3: Candidate-cost CNN used to rank globally available distance-one ZERO patterns. The scalar output estimates reciprocal-distance replacement cost.

and seed 20260607. Complete source images define the train/validation split at image level.

Figure 3 summarizes the evaluated architecture and its scalar candidate-cost output.

Random-Forest uniform-block ablation

Uniform blocks are processed by a second reversible layer. A Random Forest classifier predicts whether a block admits a low-cost single-pixel flip from its 9×9 context, and a second forest predicts the pixel position. The safety label uses a local reciprocal-distance cost at most 0.5. Selected coordinates and positions are compactly serialized for exact extraction and restoration. The experiment shows that the classifier identifies reliable locations, while the coordinate stream defines the next compression target. The final net-aware configuration therefore assigns its bit budget to the enumerative block-mapping layer.

Both forests use 120 trees and maximum depth 14. The safety model uses minimum leaf size 2 and balanced class weights; the position model uses minimum leaf size 1. Validation is grouped by source image with three folds, seed 20260607, and a deployment probability threshold of 0.7. These bounded defaults were fixed before the net-capacity evaluation: 120 trees stabilize the grouped predictions, depth 14 limits memorization of image-specific contexts, and the 0.7 threshold favors precision when every accepted block incurs coordinate overhead. They were fixed before evaluating the reported test outcomes. This layer is an evaluated ablation, and the final payload figures focus on the enumerative block-mapping layer because the current coordinate stream outweighs the measured gross gain.

Net-capacity accounting

Gross mapping capacity is the number of embeddable symbols offered by the selected active tables before the requested application payload is truncated. The map-selection score subtracts the exact serialized side information:

$$C_{\text{net}}^{\text{map}} = C_{\text{gross}} - |A_{\text{wire}}|_2. \tag{16}$$

For distance-one mappings, the wire format ranks the sorted active-PEAK subset in the combinatorial number system. The corresponding nine possible flip positions are then represented as base-9 digits. This representation closely matches the information content of the mapping set and is more compact than independent fixed-width pairs. Mapping pairs are ordered by occurrence count, and the highest-scoring prefix is retained. When the best score is at most zero, the encoder returns the unchanged image.

For t active PEAKs $0 \leq p_1 < \cdots < p_t < 512$, the subset rank is

$$r(\mathcal{P}) = \sum_{j=1}^{t} \binom{p_j}{j}. \tag{17}$$

If $s_j \in \{0, \ldots, 8\}$ is the flipped bit position, the combined integer is

$$u = r(\mathcal{P})9^t + \sum_{j=1}^{t} s_j 9^{t-j}. \tag{18}$$

Table 2
Executable auxiliary-information model used for exact extraction and restoration.

Method	Auxiliary element	Specified in cited paper	Executable representation	Charged bits
ABM	Encoding marker	Proposed codec	raw/compressed marker byte	8
ABM	Stream identity	Proposed codec	magic and version bytes	40
ABM	Payload length q	Proposed codec	big-endian uint32	32
ABM	Active table count t	Proposed codec	big-endian uint16	16
ABM	Active PEAKs and flips	Proposed codec	byte-aligned joint state in Eq. 19	variable
Huynh–Nguyen	Peak, selected states, q	Partial	compact byte stream with zlib option	measured
Dong	Context divisor, end position, q, PF/PFR map	Partial	packed and zlib-compressed location map	measured
PPOCP	Pair level, used centers, original centers, q	Partial	varint positions plus bit-packed original centers	measured

Image dimensions are supplied by the marked-image container for ABM and checked by the decoder API. Baseline synchronization formats are treated as conservative executable formats for the common round-trip protocol.

For example, with active PEAKs $\{5, 9, 12\}$ and flip positions $(0, 4, 8)$, $r = \binom{5}{1} + \binom{9}{2} + \binom{12}{3} = 261$, the base-9 component is 44, and the combined state is $u = 261 \times 9^3 + 44 = 190313$. The wire stream contains a one-byte raw/compressed marker, a four-byte magic identifier, a one-byte format version, the 32-bit big-endian payload length q, the 16-bit big-endian table count t, and the shortest big-endian byte string capable of representing the joint state space $\binom{512}{t}9^t$. Thus the variable state is charged as

$$8\left\lceil \frac{\left\lceil \log_2\left(\binom{512}{t}9^t\right)\right\rceil}{8}\right\rceil \tag{19}$$

bits, and the fixed ABM header is 96 bits. The active PEAKs are sorted by increasing pattern index inside the combinatorial rank, while the frequency-ordered prefix determines which PEAKs are active. Image dimensions are provided by the stego container and are checked by the extraction API. The stream is a serialized plaintext synchronization channel; confidentiality and authentication, when required by an application, are orthogonal transport-layer services.

Table 2 gives the executable synchronization model used in the comparisons. For ABM the listed representation is part of the proposed codec. For baselines, the original publications specify the reversible principle and some parameters while leaving the complete independent byte stream open; the table therefore records the conservative executable representations implemented for this protocol.

3.6. Comparison with Conventional Block-Mapping

Huynh–Nguyen [4] is the closest direct 3×3 PEAK/ZERO block-mapping comparator. The comparison therefore isolates one difference: the block representation, the PEAK/ZERO idea, and Hamming-distance control are established foundations, while the evaluated change is serialization-aware activation. The same reversible mapping family is retained when its exact encoded description leaves usable mapping capacity after Equation 16. On the eight document benchmarks, the proposed implementation provides a mean gross mapping capacity of 3832 bits, embeds a mean application payload of 2874 bits at the 75% budget, and delivers 2245 net bits after 629 auxiliary bits are charged. Huynh–Nguyen report 2070 gross bits under their own protocol; this published value is contextual because their serialized auxiliary cost remains open.

3.7. Computational Complexity and Formal Properties

The following propositions summarize the asymptotic cost and recovery guarantees of the deployed embedding and extraction algorithms. The analysis uses the finite 512-state 3×3 alphabet to show that histogram, mapping, and inverse-dictionary structures remain constant with respect to image size, while image traversal dominates runtime. The final propositions establish deterministic reversibility and block non-interference.

Notation. Let $n = MN$ denote the total number of pixels and $B = \lfloor M/3 \rfloor \lfloor N/3 \rfloor$ the number of complete 3×3 blocks. Let $\mathcal{P}$ be the set of frequent (PEAK) patterns and $\mathcal{Z}$ the set of unused (ZERO) patterns. For a pattern $P_i \in \mathcal{P}$, let n_i be the number of blocks exhibiting P_i, L_i the embedding capacity associated with P_i, and $E = \sum_i n_i^{\text{emb}}$ the total number of blocks actually used for embedding.

Proposition 3.1 (Embedding Time Complexity)**.** *The time complexity of the deployed distance-one embedding algorithm is*

$$O\big(n + |\mathcal{P}_0|\, c_\theta + E\big) = O(n), \tag{20}$$

where c_θ is the fixed cost of evaluating at most nine candidates with the fixed CNN.

Proof. The algorithm first partitions the image into blocks and computes pattern indices, which requires a single pass over all pixels and thus $O(n)$ time. Histogram construction and pattern classification are performed in constant time per block, yielding $O(B) = O(n)$ complexity. For each PEAK, its at most nine distance-one neighbors are generated directly and ranked by the fixed-size CNN, requiring $O(|\mathcal{P}_0| c_\theta)$ time. Since the binary 3×3 pattern space contains only 512 states, this term is bounded independently of image size. Finally, each embedding operation consists of a constant-time table lookup and block substitution, and is executed for E blocks, contributing $O(E)$ time. Summing all terms gives the stated complexity. □

Proposition 3.2 (Embedding Space Complexity)**.** *The space complexity of the embedding algorithm is*

$$O(n + q). \tag{21}$$

Proof. The algorithm stores the original and modified images, requiring $O(n)$ space. The histogram, mapping tables, and inverse dictionary contain at most 512 states and therefore require constant space with respect to image size. The payload buffer requires $O(q)$ space. □

Proposition 3.3 (Optimized Extraction Time Complexity)**.** *The optimized extraction algorithm runs in*

$$O(n) \tag{22}$$

time.

Proof. Extraction scans the stego image once to reconstruct blocks and compute pattern indices, which requires $O(n)$ time. The inverse dictionary has at most 512 entries. Each complete block is processed at most once in deterministic raster order with constant-time lookup, and extraction stops after the declared q bits. □

Proposition 3.4 (Extraction Space Complexity)**.** *The space complexity of the optimized extraction algorithm is*

$$O(n + q). \tag{23}$$

Proof. During extraction, memory is required for the stego image, the recovered image, and the inverse dictionary constructed from the mapping tables. The inverse dictionary is bounded by the 512-state pattern alphabet, and the message buffer stores exactly q recovered bits. □

Proposition 3.5 (Deterministic Reversibility)**.** *Given an unmodified stego image and its valid auxiliary stream, the proposed embedding and extraction procedures recover the payload and cover exactly.*

Proof. Each table maps 0 to its PEAK and 1 to one absent distance-one pattern. Assigned ZEROs are globally disjoint and were absent from the cover, so the union of table ranges is injective. During extraction, the inverse dictionary therefore maps every active stego pattern to one bit and one original PEAK. Since blocks are non-overlapping, modified at most once, and processed in a deterministic traversal order, the declared payload length gives a unique stopping point. Unused complete blocks and incomplete border pixels are never modified. Hence both the embedded message and original image are reconstructed exactly. □

Proposition 3.6 (Non-interference of Distinct Blocks)**.** *Conditional on the fixed mapping tables, an embedding modification is confined to its current non-overlapping block and preserves the pixel state of every other block.*

Proof. Each operation uses the pattern-specific table for the current block and writes only the nine pixels in that block. The raster order determines which payload symbol is consumed next, while the pixel update remains spatially confined to the current block. Distinct complete blocks are disjoint, so one block update leaves all other blocks unchanged. □

4. Experimental Evaluation

4.1. Datasets, Splits, and Reproducibility

The evaluation has three complementary validation layers. The eight standard 512×512 binary document images used by Huynh–Nguyen [4] support direct validation against the block-mapping literature; their diversity is shown in Figure 4. BOSSbase [27] provides 10,000 grayscale images, from which seed 20260608 selects 100 profile images and 100 disjoint test images. The profile split supplies the image-level statistics needed by the conservative Dong implementation. BOSSbase test images are binarized at threshold 128. The CNN ranker trained on the document images is transferred directly to BOSSbase, preventing test-set adaptation. Finally, all 199 scanned documents in FUNSD form an external real-document validation set. The committed CNN ranker is frozen for this experiment: no training, fine-tuning, threshold fitting, or other model tuning is performed on FUNSD. Each FUNSD page is evaluated under both fixed threshold 128 and Otsu binarization.

The four-method BOSSbase protocol embeds identical pseudorandom messages at targets 64, 128, 256, and 512 bits. A run counts as available after exact message extraction and bitwise cover restoration. Huynh–Nguyen uses the published threshold $T = 5$. Dong evaluates all context divisors $l \in \{1, \ldots, 10\}$ and retains the exact minimum-DRD result. PPOCP and Dong use conservative serialized synchronization or location information where the publications leave the independent wire representation open. Accordingly, net-payload conclusions apply to the executable wire formats defined in this study, while DRD and availability provide a comparison less sensitive to byte-stream conventions.

The FUNSD protocol is deliberately narrower and tests transfer rather than a new four-method ranking. At each target $q \in \{64, 128, 256, 512\}$, it compares A1, which activates all feasible tables and then charges their exact wire cost, with the proposed A2 serialization-aware prefix selection. The same frozen CNN weights and seed 20260608 are used throughout. Availability, auxiliary bits, delivered net payload, DRD, positive-net fraction, and exact message/cover round-trip recovery are recorded per image; paired A1/A2 summaries are computed only where both variants are available.

4.2. Metrics and Statistical Protocol

The requested application payload is the message length $q = |M|$. Gross mapping capacity is C_{gross}, auxiliary bits are the exact serialized wire length $C_{\text{aux}} = |A_{\text{wire}}|_2$, and delivered net payload is $P_{\text{net}} = q - C_{\text{aux}}$. The main text reports DRD [15] as the primary distortion metric; changed-pixel and PSNR outputs are retained in the reproducibility artifacts. Availability is the fraction of test images that carry the target and complete the exact round trip.

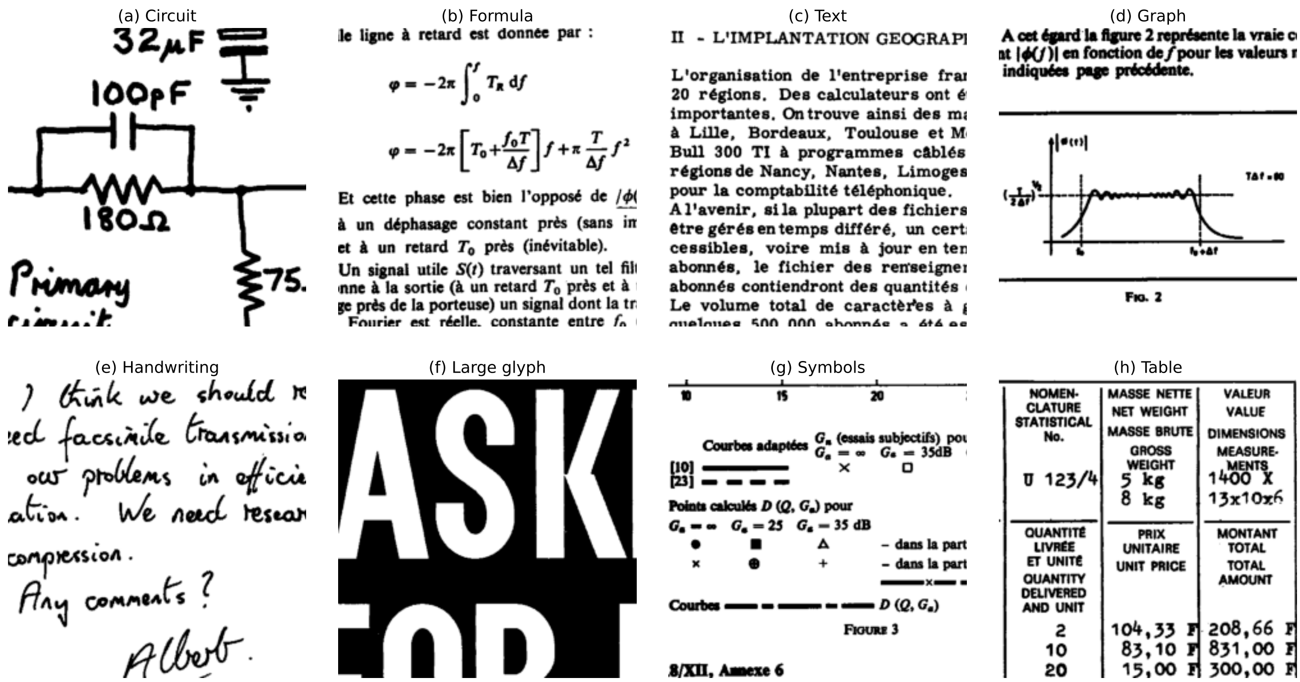

Figure 4: Eight binary document images used for block-mapping validation.

Primary per-payload comparisons use common available images. A secondary fixed cohort contains the 46 images on which all four methods are available at all four payloads. ABM is paired with each baseline for net bits and DRD using the Wilcoxon signed-rank test and Holm correction within each payload. Mean net payload is accompanied by a 95% percentile bootstrap interval with 10,000 image-level resamples and seed 20260608. Median DRD uses the analogous 95% percentile bootstrap interval computed from the same image-level resamples. Runtime includes embedding, serialization, extraction, and verification. Memory is peak process RSS minus pre-call RSS, sampled every 2 ms. Experiments used Python 3.13.5, NumPy 2.2.6, PyTorch 2.8.0 CPU, scikit-learn 1.7.2, and SciPy 1.17.1 on 64-bit Windows with 12 logical processors.

4.3. Serialization-Aware Activation Ablation

Table 3 isolates the central algorithmic decision. All variants use the same CNN distance-one candidate assignment; only active table selection and exact wire charging differ. A0 is the gross-only selection reference: it activates all feasible tables without charging the wire in the selection decision; the table separately reports gross mapping capacity, the 75% application payload, and the post-hoc net payload after exact wire charging. A1 charges the wire after activating all feasible tables. A2 is the proposed serialization-aware activation, which selects the frequency-ordered prefix that maximizes Equation 16. Relative to A1, A2 reduces auxiliary information by 180 bits on average and raises mean net payload by 105.9 bits (4.95%) while preserving exact reversibility on all eight document images.

4.4. Candidate Ranking and CNN Architecture

Table 4 compares the learned ranker with simple non-learning alternatives under the same admissible distance-one ZERO set and common payload per image. The CNN is close to the direct DRD-cost ranker and improves mean DRD by about 5% relative to Hamming-index ordering. The reversible mechanism remains the deterministic, globally injective PEAK/ZERO substitution defined in Section 3.

Table 5 gives the controlled CNN architecture diagnostic. All configurations use 20 epochs, AdamW with learning rate 10^{-3}, Smooth-L1 loss, batch size 128, the same two-image train/validation split, and deterministic seeds. The deployed two-layer w16 9×9 model is retained as a compact compromise: it is close to the lowest-MAE wider model, faster at inference, and keeps held-out placement DRD within the same practical range.

4.5. Serialization Sensitivity for Baselines

The original descriptions of PPOCP, Dong, and Huynh–Nguyen specify the reversible principles while leaving the complete standalone byte stream partly unspecified. The common protocol therefore uses conservative executable synchronization. We also ran a header-credit sensitivity check: for each baseline, the method-identifying header, version field, and image-shape fields are removed from the charged cost, giving a 104-bit credit while retaining the payload-critical maps, states, locations, and payload length. With this credited accounting, ABM retains the highest mean net payload on every all-method matched cohort: −57 vs. −120 bits at 64, −7 vs. −72 bits at 128, 93 vs. 23 bits at 256, and 309 vs. 200 bits at 512 for the best credited baseline. This

Table 3
Central ablation of serialization-aware mapping activation on the eight document images. Values are means in bits except DRD; the post-hoc value in A0 is the net payload obtained when the exact wire is charged after the gross-only selection.

Variant	Active-table selection	Wire charged	Gross cap.	Payload	Aux.	Net payload	Mean DRD
A0	all feasible; gross reported	Gross only	3932	2948	0	2948 (2139 post hoc)	0.771
A1	all feasible; wire charged	Charged	3932	2948	809	2139	0.778
A2 proposed	serialized-net prefix	Charged	3832	2874	629	2245	0.761

Table 4
Ranker ablation with identical distance-one candidates and common payloads.

Ranker	Mean net	Mean DRD	Round trip
Hamming index	2141	0.822	Exact
Direct DRD cost	2139	0.778	Exact
Transition preserving	2140	0.829	Exact
Connectivity aware	2140	0.820	Exact
CNN	2137	0.780	Exact

fixed-header variation leaves the useful-capacity ordering unchanged for the tested executable representations.

4.6. Machine-Learning Validation

Image-wise leave-one-out (LOSO) validation trains eight independent CNNs, each on seven document images and tests it on the remaining complete image. Across 13,808 held-out candidates, pooled mean absolute error is 0.0367 and pooled Pearson correlation is 0.9893; per-image correlations range from 0.9861 to 0.9939. A separate LOSO placement experiment applies each fold model to its held-out image at an identical 75% distance-one payload. Mean DRD decreases from 0.8054 with deterministic Hamming-index ordering to 0.7663 with CNN ordering, a 4.85% reduction. All eight images improve, with per-image reductions from 2.83% to 7.94%, while every message and cover is recovered exactly. The separate two-image architecture diagnostic in Table 5 evaluates 4323 held-out candidates under the fixed 20-epoch protocol. Thus, regression accuracy and operational DRD impact are evaluated separately, with fold-level regression and placement results provided with the reproducibility artifacts.

The uniform-block Random Forest is evaluated by source-image-grouped cross-validation on 14,798 samples. It reaches ROC AUC 0.995, precision 0.897, recall 0.970, and flip-position accuracy 0.971. In the combined reversible pipeline, the layer adds 181.4 gross bits per document image on average, while its current coordinate representation uses 2691.0 auxiliary bits. The resulting combined net payload is therefore −264.6 bits on average, compared with 2245.0 bits for the base enumerative block-mapping layer. This identifies coordinate coding as a concrete optimization opportunity and keeps the Random Forest layer as a diagnostic component.

4.7. Document-Image and External Net Capacity

Table 6 reports the integrated CNN-ranked mapping layer at a 75% payload budget. Auxiliary sizes are measured from serialized bytes, and all eight messages and covers are recovered exactly.

On the 100 BOSSbase test images, net-aware table selection gives a mean embedded payload of 669.6 bits, 377.1 auxiliary bits, and 292.5 delivered net bits. Median net capacity is 159 bits, 75 images are net positive, mean DRD is 0.300, and every evaluated message and cover is recovered exactly. Disabling net-aware table selection on the same sample yields a mean embedded payload of 832.5 bits, 731.4 auxiliary bits, and 101.1 delivered net bits, showing that the active-prefix decision also improves the external transfer setting.

4.8. Frozen External Validation on FUNSD

Table 7 reports the frozen external validation on all 199 FUNSD scanned documents. The existing CNN ranker is used without retraining or tuning. Two independently generated binary inputs are tested per source page: fixed threshold 128 and Otsu. A1 activates all feasible tables and charges the resulting wire, whereas A2 applies the proposed serialization-aware prefix selection. Means are computed over available runs for each variant.

Every available FUNSD run recovers the payload exactly and restores the binary cover bit-for-bit. On paired available images, A2 is never worse than A1 in net payload: the fraction $P_{\text{net}}^{A2} \geq P_{\text{net}}^{A1}$ is 1.0 at every payload under both binarizations. For fixed-128, mean A2 net gains (and equal auxiliary savings) are 240.3, 240.3, 240.3, and 241.4 bits at 64, 128, 256, and 512 bits, respectively; for Otsu they are 238.9, 238.9, 238.9, and 239.8 bits. The median gain is 232 bits in every condition. The associated mean DRD differences $\text{DRD}_{A2} - \text{DRD}_{A1}$ remain small, ranging from 6.66×10^{-5} to 7.99×10^{-4} across the eight binarization/payload conditions.

The external result therefore supports two transfer claims that are distinct from net-efficiency claims. First, exact reversible operation generalizes to a large real scanned-document corpus without retraining. Second, the proposed serialization-aware activation continues to reduce synchronization overhead on every paired operating point. By contrast, the delivered net payload remains metadata-dominated throughout the tested FUNSD range: mean net payload is negative through 512 bits, and the largest observed positive-net fraction is 4/198 (2.02%). Thus FUNSD validates reversibility and the A2 activation benefit, but it does not

Table 5
CNN architecture diagnostic on the two-image detailed split. Inference time is measured on 4323 held-out candidates. Held-out DRD is the mean DRD after model-ranked placement on the diagnostic held-out images. All configurations share the reversible mapping and auxiliary stream.

Configuration	Params	Train s	Inf. μs	MAE	Pearson	Held-out DRD
1 conv., 9×9, w24	14625	6.97	7.18	0.0716	0.962	0.848
2 conv., 9×9, w24	38865	23.62	27.19	0.0339	0.991	0.830
3 conv., 9×9, w24	44073	27.80	38.21	0.0347	0.991	0.848
2 conv., 7×7, w24	38865	18.11	19.51	0.0383	0.989	0.845
2 conv., 11×11, w24	38865	27.77	56.13	0.0485	0.985	0.825
2 conv., 9×9, w16 (deployed)	23649	15.16	24.86	0.0379	0.990	0.839
2 conv., 9×9, w32	56385	27.27	40.62	0.0303	0.992	0.847

Table 6
Net-aware results on the eight binary document images.

Image	Payload	Aux.	Net	DRD
circuit-2	1576	592	984	0.583
formula-5	3692	672	3020	0.900
graph1-5	1978	632	1346	0.662
handwr2-8	2351	632	1719	0.768
large-8	1138	488	650	0.616
symbol-6	2367	672	1695	0.897
table1-3	4161	656	3505	0.789
french-4	5729	688	5041	0.864
Average	**2874**	**629**	**2245**	**0.760**

establish positive net efficiency for scanned documents at these payload sizes.

4.9. Multi-Payload Common-Protocol Results

Table 8 and Figure 5 summarize the all-method matched BOSSbase subsets. ABM moves from metadata-dominated operation at 64 bits to a near break-even 128-bit point: the payload-specific matched cohort has a −7-bit mean, while the fixed 46-image cohort has an 8-bit mean. It reaches positive mean net payload at 256 and 512 bits on these BOSSbase cohorts. PPOCP defines the lowest-DRD curve, while ABM defines the highest-net curve.

Table 9 preserves the ordering observed in the payload-specific matched analysis. It also shows that ABM reaches a positive 128-bit mean on the fixed cohort.

Figure 6 complements the table with availability and the net-bits–DRD operating curves. PPOCP carries every target on all test images. ABM availability progresses from 89% at 64 bits to 70% at 512 bits, matching Dong at the highest target and remaining above Huynh–Nguyen. The right panel is descriptive because the methods span different DRD ranges; Table 9 provides the controlled cross-payload comparison.

4.10. Paired Significance and Binarization Strata

At each payload, ABM's paired net advantage over Dong, Huynh–Nguyen, and PPOCP remains significant after Holm correction. At 256 bits, adjusted p-values are below 6.7×10^{-13}, 3.1×10^{-13}, and 2.2×10^{-14}, respectively. Median ABM net gains are 160, 1752, and 3016 bits, with paired rank-biserial correlations of 0.974, 1.000, and 1.000. DRD tests place ABM below Dong and above the perceptually focused PPOCP and Huynh–Nguyen curves, yielding a two-objective interpretation.

Binarization suitability is defined before embedding: threshold 128 must agree with Otsu on at least 95% of pixels, and foreground fraction must lie in $[0.05, 0.95]$. Twenty-two test images satisfy this criterion. As shown in Figure 7, ABM is available on all 22 through 256 bits, while net-positive behavior at 256 and 512 bits is present in both strata. Suitability therefore serves as an applicability indicator, with net performance reported separately in both strata.

4.11. Runtime and Memory

Figure 8 and Table 10 report the 256-bit resource benchmark on ten protocol images. PPOCP provides the shortest runtime, Huynh–Nguyen the smallest incremental memory profile, and ABM requires a median runtime of 3.56 s and median incremental peak RSS of 2.56 MiB. Dong's complete ten-divisor optimization quantifies the computational cost of per-image context search.

Table 11 decomposes the proposed method on the eight document images at the 75% payload budget. Candidate generation, including CNN-ranked distance-one assignment, and embedding dominate runtime; the enumerative serialization itself is negligible.

4.12. Grouped Steganalysis

The security evaluation uses foreground fraction, horizontal and vertical transition rates, and normalized 2×2 and 3×3 pattern histograms. Five-fold GroupKFold keeps each cover and its stego in the same fold. Both a standardized logistic model and a 500-tree Random Forest are evaluated.

Figure 9 visualizes the complete AUC curves, and Table 12 reports the logistic detector. PPOCP provides the most resistant curve from 128 to 512 bits. ABM's AUC increases with useful payload, making statistical footprint the principal signal for future optimization. The Random Forest gives ABM AUC values 0.623, 0.691, 0.769, and 0.851, confirming the same payload trend with a different model.

Table 7
Frozen external validation on 199 FUNSD scanned documents. Auxiliary and net values are mean bits over available runs; DRD and positive-net counts refer to A2. Availability is shown as available/199.

Binarization	q	A1 avail.	A2 avail.	A1 aux.	A2 aux.	A1 net	A2 net	A2 DRD	A2 net>0
Fixed 128	64	199/199	199/199	965.7	725.4	−901.7	−661.4	0.01049	0/199
Fixed 128	128	199/199	199/199	965.7	725.4	−837.7	−597.4	0.02082	0/199
Fixed 128	256	199/199	199/199	965.7	725.4	−709.7	−469.4	0.04124	1/199
Fixed 128	512	198/199	198/199	969.3	727.8	−457.3	−215.8	0.08255	4/198
Otsu	64	199/199	198/199	985.9	750.9	−921.9	−686.9	0.01015	0/198
Otsu	128	199/199	198/199	985.9	750.9	−857.9	−622.9	0.02008	0/198
Otsu	256	198/199	198/199	989.8	750.9	−733.8	−494.9	0.03981	0/198
Otsu	512	197/199	197/199	993.2	753.4	−481.2	−241.4	0.07928	3/197

Table 8
Mean net bits and median DRD on all-method matched images.

		Proposed ABM		Dong		Huynh–Nguyen		PPOCP	
Target	Matched	Net	DRD	Net	DRD	Net	DRD	Net	DRD
64	86	−57	0.041	−224	0.056	−2200	0.035	−997	0.023
128	83	−7	0.080	−176	0.108	−2096	0.070	−1653	0.041
256	69	**93**	0.135	−81	0.200	−1812	0.117	−2856	0.064
512	46	**309**	0.176	96	0.332	−1191	0.178	−4822	0.097

Table 9
Fixed-cohort sensitivity analysis on the 46 images available for every method and payload. Values are mean net bits with 95% bootstrap intervals; DRD is the cohort median.

Target	Method	Mean net bits (95% CI)	Median DRD
64	Proposed ABM	−48.5 [−49.0, −48.0]	0.022
	Dong / Huynh–Nguyen / PPOCP	−226.3 [−229.6, −223.0] / −1644.7 [−1854.3, −1447.5] / −1028.0 [−1063.3, −990.1]	0.029 / 0.024 / 0.012
128	Proposed ABM	8.0 [5.7, 10.1]	0.043
	Dong / Huynh–Nguyen / PPOCP	−178.3 [−187.8, −169.9] / −1582.8 [−1792.4, −1386.8] / −1707.8 [−1767.5, −1643.0]	0.066 / 0.044 / 0.025
256	Proposed ABM	113.9 [108.3, 119.0]	0.087
	Dong / Huynh–Nguyen / PPOCP	−83.8 [−105.6, −65.7] / −1456.3 [−1656.2, −1260.9] / −2954.1 [−3060.2, −2835.8]	0.132 / 0.084 / 0.050
512	Proposed ABM	309.0 [289.7, 325.9]	0.176
	Dong / Huynh–Nguyen / PPOCP	95.8 [55.1, 131.0] / −1190.6 [−1397.6, −996.7] / −4821.7 [−5025.6, −4607.1]	0.332 / 0.178 / 0.097

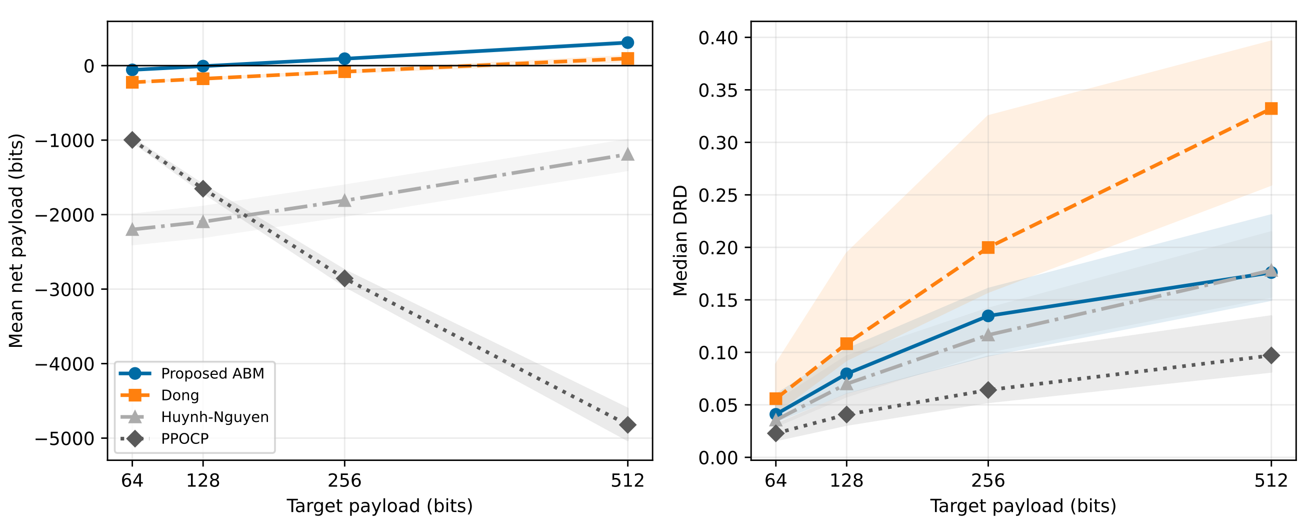

Figure 5: Payload scaling on all-method matched images: mean net payload (left) and median DRD (right), with 95% bootstrap confidence bands.

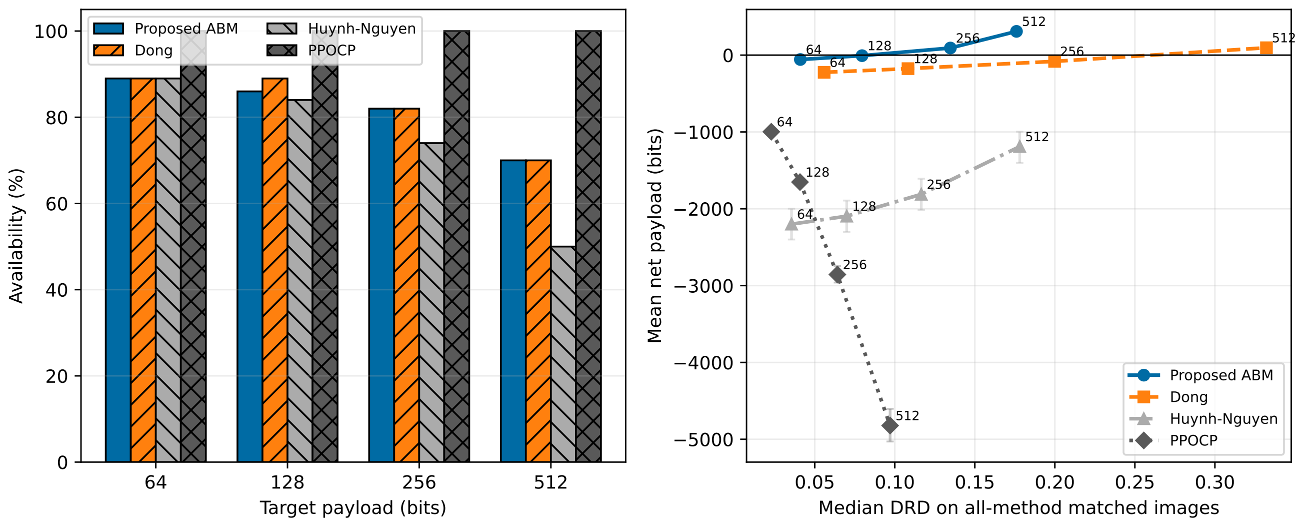

Figure 6: Complementary views of payload feasibility and useful-capacity efficiency. Error bars denote 95% bootstrap confidence intervals for mean net payload; numbers on the right panel denote target payloads in bits.

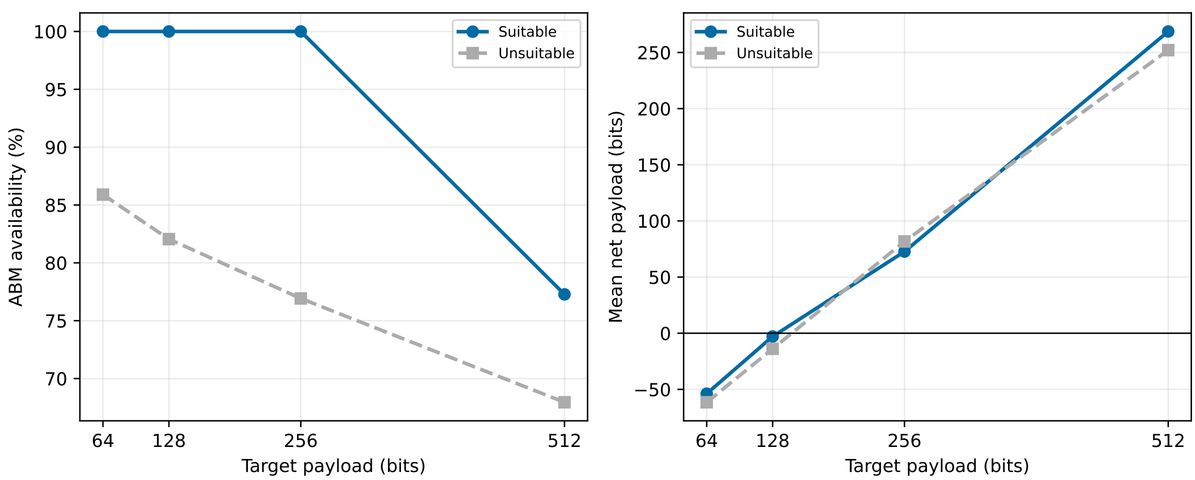

Figure 7: ABM availability and mean net payload after stratification by fixed-threshold/Otsu agreement.

A document-image payload sweep explains this trend. When the ABM payload fraction increases from 25% to 100%, mean net payload rises from 329 to 3203 bits, but the mean Jensen–Shannon divergence between cover and marked 3×3 pattern histograms also rises from 0.0087 to 0.0409. Mean horizontal and vertical transition-rate displacements rise from 0.0032/0.0032 to 0.0127/0.0128. The steganalysis features therefore respond to the same PEAK/ZERO redistribution that produces useful capacity: repeated PEAK patterns are depleted and distance-one ZERO patterns are populated. High payload is therefore best interpreted as capacity-oriented reversible annotation in access-controlled channels. Figure 10 gives the qualitative counterpart: residual locations show how each method distributes its changes under the same 512-bit target on table1-3.

Table 10
Median resource profile at a 256-bit target.

Method	Available	Time (s)	Peak RSS increase (MiB)
Proposed ABM	9/10	3.56	2.56
Dong	9/10	16.97	0.05
Huynh–Nguyen	7/10	1.51	0.02
PPOCP	10/10	0.16	0.70

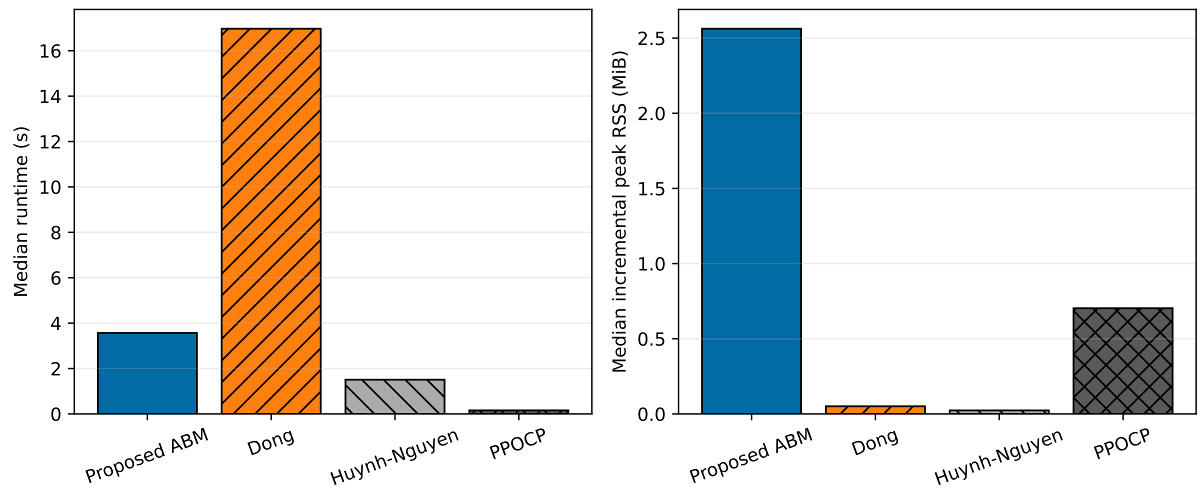

Figure 8: Median wall-clock runtime and incremental peak RSS at 256 bits.

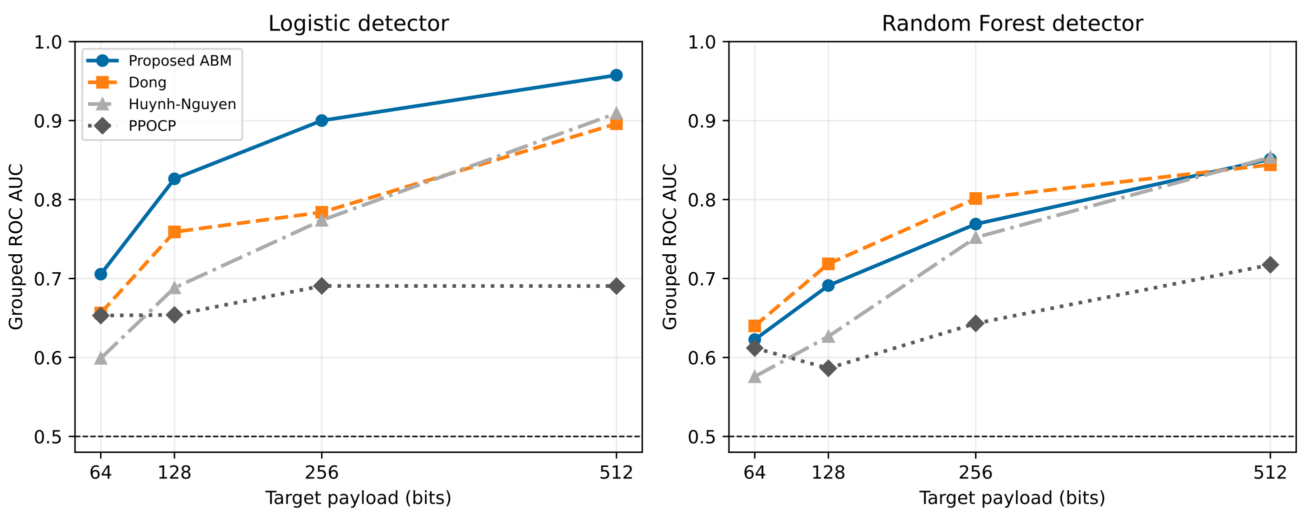

Figure 9: Grouped steganalysis across payloads with logistic (left) and Random Forest (right) detectors. The dashed line denotes chance-level AUC.

Table 11
ABM runtime decomposition by module on document images.

Module	Median time (s)
Candidate generation and ranking	1.1601
Serialized-prefix optimization	0.5268
Embedding	1.3893
Serialization	0.00025
Extraction	0.3213
Round-trip verification	0.00012
End-to-end total	3.4409

The end-to-end timer includes minor orchestration overhead outside the module timers.

5. Discussion

5.1. Positioning Across Practical Objectives

The experiments separate four objectives that are frequently merged in RDH comparisons. In the BOSSbase common protocol, ABM provides the highest net-payload curve and crosses into mean-positive net payload between 128 and 256 bits on the payload-specific matched cohorts. In perceptual quality, PPOCP defines the lowest median DRD, followed closely by Huynh–Nguyen at the higher target. In applicability, PPOCP has universal target availability, while ABM combines high availability with compact side information. In computational cost, ABM remains practical and is substantially faster than the complete Dong parameter search.

This decomposition supports a specific, protocol-bounded position: among the four evaluated implementations, ABM provides the highest net payload after serialized metadata, whereas PPOCP provides the lowest-distortion and most favorable detectability profile. This conclusion rests on exact round trips, multi-payload curves, a fixed-cohort sensitivity

Table 12
Grouped logistic-detector ROC AUC on matched cover/stego pairs.

Target	Pairs	Proposed ABM	Dong	Huynh–Nguyen	PPOCP
64	86	0.706	0.656	**0.599**	0.653
128	83	0.826	0.759	0.688	**0.654**
256	69	0.900	0.784	0.774	**0.691**
512	46	0.957	0.896	0.909	**0.690**

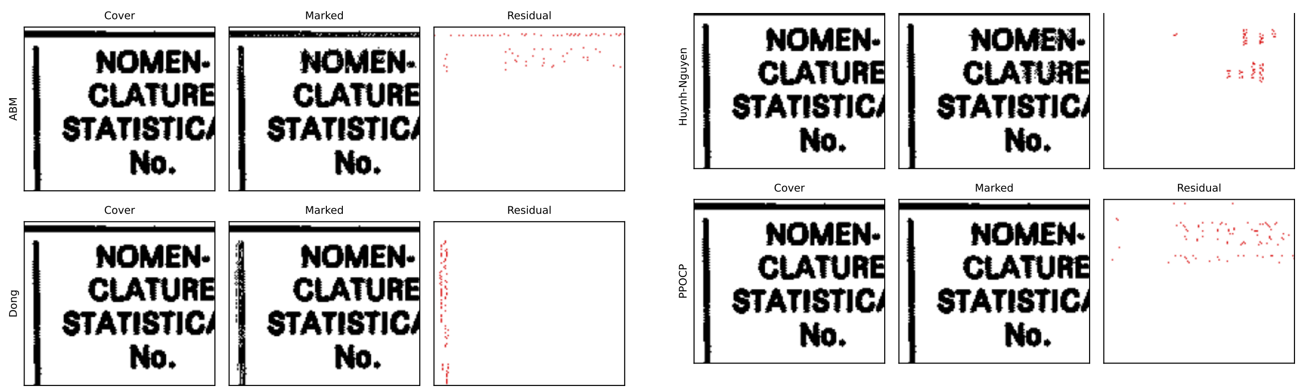

Figure 10: Cover, marked, and residual zooms on table1-3 at a 512-bit target, split into two panels to fit the two-column layout: ABM and Dong appear on the left, and Huynh–Nguyen and PPOCP on the right. Residual pixels are shown in red on the marked crop to make modification locations visible under the same payload.

analysis, and paired tests. Because synchronization formats materially affect net payload, the result is framed as a protocol-bounded comparison of the evaluated implementations.

FUNSD separates external-document transfer from net efficiency. The frozen codec remains exactly reversible on every available run and A2 reduces auxiliary cost on every paired operating point under both binarizations, which materially strengthens the evidence for transfer of the reversible mechanism and the serialization-aware activation rule. At the same time, mean delivered net payload remains negative through 512 bits and positive-net cases are rare. Those values characterize the tested low-to-moderate-payload operating regime on scanned documents; they should not be generalized into a claim of positive net efficiency on FUNSD.

5.2. Security-Aware Optimization Direction

The measured AUC curves and content-adaptive detectability principles [28] suggest a direct route for strengthening the method. The current CNN predicts reciprocal-distance cost, whereas the steganalysis features respond to shifts in local pattern distributions. A security-aware ranker can optimize

$$\mathcal{L}(Z_j) = \beta\,\widehat{\mathrm{DRD}}(Z_j) + \gamma\,\Delta_{\mathrm{hist}}(Z_j) + \eta\,\Delta_{\mathrm{trans}}(Z_j), \quad (24)$$

where Δ_{hist} measures the change in local pattern histograms and Δ_{trans} measures transition-rate displacement. The coefficients can be selected on a Pareto front of net bits, DRD, and grouped detector AUC. This extension builds directly on the validated ranker and preserves the exact reversible wire format.

The present measurements define the operating envelope. The tested settings are interpreted as detectability-characterized annotation operating points rather than covert-security thresholds. At 256 and 512 bits, logistic AUC values of 0.900 and 0.957 place the current ABM configuration in a capacity-oriented reversible-annotation regime, despite its positive net payload and exact reversibility. At 64 bits, AUC 0.706 reflects lower detectability, although metadata overhead dominates useful capacity. PPOCP remains the preferred evaluated method for security-prioritized operation, whereas ABM addresses applications in which useful reversible capacity is primary and the stego channel itself is access-controlled.

The RF uniform-block experiment points to a second optimization path: coordinate coding can use enumerative subset coding conditioned on the deterministic block order. Its grouped classification metrics show that location selection is reliable; compressing the selected subset targets the measured source of overhead.

5.3. Experimental Scope

The eight document images provide continuity with the binary block-mapping literature, the disjoint BOSSbase protocol adds 100 external test images with a four-method common evaluation, and FUNSD adds a separate frozen external test on all 199 real scanned documents under two binarization rules. The latter removes the earlier absence of a sizeable real scanned-document benchmark as a primary

Table 13
Channel stress test for ABM with the same auxiliary stream.

Channel	Stego	Message	Cover
PNG	Yes	Yes	Yes
TIFF	Yes	Yes	Yes
JPEG q=5	Altered	Mismatch	Mismatch
One-pixel noise	Altered	Mismatch	Mismatch

limitation of the evidence: exact recovery and the A2-over-A1 serialization benefit are now observed without retraining on that corpus.

Conditions of validity. The exact-reversibility guarantee is conditional on a matched stego image and its valid auxiliary stream being preserved through a lossless storage or transmission path. The encoder may reject a requested payload when the binary cover does not offer sufficient active mapping capacity; such rejection is an availability outcome, not a recovery failure. A stress test on table1-3 at 256 bits confirms exact recovery after PNG and TIFF round trips, while JPEG recompression at quality 5 and a one-pixel perturbation produce mismatched stego/auxiliary pairs (Table 13). Applications requiring lossy, noisy, or geometrically transformed channels therefore require an external synchronization or error-control layer before the present exact recovery contract applies.

Operating characteristics. Auxiliary overhead is amortized differently across image distributions and payload sizes. On BOSSbase matched cohorts, mean net payload becomes positive between 128 and 256 bits, whereas on FUNSD the fixed and mapping-dependent serialization cost dominates through the tested 512-bit target. FUNSD availability remains 98.99–100% depending on binarization, activation, and payload, and every available run is exact. DRD and detectability also increase with payload. These are measured properties of the operating envelope rather than failures of the reversible contract.

Remaining limitations. Three limitations remain material. First, the high logistic AUC at larger BOSSbase payloads limits claims for covert or security-prioritized use under the tested detector. Second, cross-method net-payload rankings remain partly protocol-dependent because the baseline publications do not completely specify standalone serialized synchronization formats. Third, although FUNSD provides substantially stronger real-document evidence, it is one external scanned-form corpus; generalization across additional independent document collections, scanner pipelines, languages, and acquisition conditions remains unverified. The present evidence also does not establish positive-net efficiency for real scanned FUNSD documents at payloads up to 512 bits.

6. Conclusion

This work develops and evaluates serialization-aware activation for net-capacity-driven PEAK/ZERO block mapping in binary-image RDH. The enumerative codec makes the active mapping description executable and charged; the CNN ranker improves candidate placement on held-out images as a secondary cost-ordering component. The complete system delivers 2245 net bits on average on eight document images and 292.5 net bits on 100 BOSSbase images. A frozen external validation on all 199 FUNSD scanned documents further shows that every available run restores both message and binary cover exactly without training or tuning on FUNSD.

On FUNSD, serialization-aware A2 activation reduces auxiliary cost by about 239–241 bits relative to A1 on paired available images and is never worse in net payload under either fixed-128 or Otsu binarization. This is strong evidence that exact reversibility and the activation benefit transfer to real scanned documents. It is not evidence of positive net efficiency in the tested FUNSD payload range: mean net payload remains negative through 512 bits and the largest positive-net fraction is 2.02%.

Under the BOSSbase common protocol, ABM obtains mean net payloads of −57, −7, 93, and 309 bits at targets 64, 128, 256, and 512 bits on matched images. Its paired net advantage is significant against every baseline after Holm correction; at 256 bits, all adjusted p-values are below 6.7×10^{-13}. Together, availability, binarization, runtime, memory, and two-classifier steganalysis make this result a reproducible practical profile beyond a single capacity number.

Under the stated implementations and wire-level accounting, the combined evidence identifies ABM as the highest useful-capacity curve in the evaluated common protocol, while PPOCP retains the lowest measured distortion and most favorable detectability profile. The validity of exact recovery is conditional on matched stego/auxiliary data over a lossless path; metadata-dominated low-payload behavior is an operating characteristic; and the remaining true limitations concern high-payload detectability, protocol dependence of baseline serialization, and external breadth beyond the single FUNSD document corpus.

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CRediT authorship contribution statement

Stéphane Gaël R. Ekodeck: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Vivien Beyala Kamgang:** Conceptualization, Methodology, Writing – review and editing. **Serge Alain Ebélé:** Conceptualization, Methodology, Validation, Writing – review and editing. **Julius Marcellin Nkenlifack:** Supervision, Project administration, Writing – review and editing.

Declarations

Ethics approval. This study does not involve human participants, animals, personal data, or clinical interventions. It uses publicly available image data and offline computational experiments.

Competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability. BOSSbase is publicly available from its original distributor at `https://dde.binghamton.edu/download/ImageDB/BOSSbase_1.01.zip` under its applicable terms. FUNSD is publicly available from the dataset provider at `https://guillaumejaume.github.io/FUNSD/dataset.zip`. The eight document images used for block-mapping validation, derived aggregate results, per-image evaluation artefacts, and the fixed-128/Otsu FUNSD external-validation summaries and paired/per-image result tables accompany the reproducibility package.

Code availability. The reproducibility package contains the ABM implementation, independently implemented comparison methods, evaluation scripts, tests, trained model weights, CSV/JSON result tables, plotting code, and exact random seeds. It is publicly available at `https://github.com/EkodeckStephane/net-aware-block-mapping-rdh`.

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The following red blocks collect deleted manuscript material whose inline placement would otherwise break LaTeX structural environments.

```
Deleted block 1
This paper presents a net-aware learned block-mapping framework for
exactly
reversible data hiding in binary images. It combines Gaussian-process
exploration, a convolutional ranker for Hamming-distance-one substitutions,
and an enumerative auxiliary codec that selects mapping pairs by useful
payload after metadata. Eight-fold leave-one-image-out validation gives
pooled
Pearson correlation 0.989 and mean absolute error 0.0367 on 13,808
candidates.
A separate held-out placement evaluation reduces mean DRD from 0.8054
to
0.7663 (4.85\%), improving every image. On eight $512\times512$ document
images, the integrated mapping layer carries 2874 gross and 2245 net
bits on
average at mean DRD 0.760. On 100 disjoint thresholded BOSSbase test
images, it
carries 516.1 net bits on average. PPOCP, Huynh-Nguyen, Dong, and the
proposed
method are evaluated with identical messages, payload targets,
distortion
computation, reversibility checks, and auxiliary-bit accounting at 64,
128,
256, and 512 bits. On all-method matched images, the proposed method
obtains
mean net payloads of $-57$, $-7$, $93$, and $309$ bits, respectively,
with a
significant paired advantage over every baseline after Holm correction.
At
256 bits, median runtime is 3.56 s and median incremental peak RSS is
2.56 MiB. Grouped steganalysis quantifies the security-payload frontier;
the
logistic AUC rises from 0.706 at 64 bits to 0.957 at 512 bits. Every
available
run recovers both message and cover exactly. The resulting contribution
is a
reproducible benchmark and the highest observed net-payload operating
point
among the evaluated implementations.
Deleted block 2
Reversible data hiding \sep Binary images \sep Block mapping \sep Net
capacity
\sep Machine learning \sep Steganalysis
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\cite{cryptographie}, whereas steganography conceals its presence
\cite{generalite}. Reversible data hiding (RDH) combines these objectives
with
Deleted block 4
Binary images provide only two pixel states, so every modification can
alter
Deleted block 5
Ren2022Sliding,Dong2020Adaptive,Zhang2022Magnification,Yin2020PPOCP,
23,Ren2019Encrypted,Feng2015Secure,69}.  Their  practical  performance
depends
Deleted block 6
The literature offers complementary operating points. Magnification-based
embedding provides low distortion at increased image size
\cite{Zhang2022Magnification}; PPOCP
Deleted block 7
the same messages, images, and distortion computation.
Deleted block 8
The proposed framework begins with adaptive block mapping and then makes
the
complete transmission cost explicit. Gaussian-process Bayesian optimization
characterizes the threshold trade-off, a convolutional neural network
ranks
distance-one  ZERO  candidates  by  predicted  perceptual  cost,  and  an
enumerative
codec compresses the active PEAK set and flip positions. A Random Forest
uniform-block  layer  is  also  implemented  and  evaluated,  providing
measured
evidence about when added gross capacity is outweighed by synchronization
metadata. Gaussian-process threshold exploration and the Random Forest
agent
are therefore reported as diagnostic ablations rather than components
of the
deployed distance-one codec.
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\item A methodological net-capacity criterion,
$C_{\mathrm{net}}=C_{\mathrm{gross}}-|A_{\mathrm{wire}}|_2$, applied
systematically to reversible block mapping through CNN-ranked
distance-one substitutions, enumerative auxiliary coding, and table-prefix
selection by useful payload.
\item A formal wire-level treatment of payload length, globally injective
PEAK/ZERO assignment, border preservation, and exact restoration.
\item Independently implemented conservative versions of PPOCP,
Huynh-Nguyen, and Dong under a common-message protocol at four payloads
on a disjoint BOSSbase test split.
\item A multi-objective evaluation covering net payload, DRD, availability,
paired significance, runtime, memory, and grouped steganalysis.
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Reversible data hiding (RDH) in binary images draws on four research
traditions  that  have  advanced  substantially  since  2018:  modern
high-capacity general-image RDH, binary-image-specific pattern manipulation,
```

deep-learning-based steganalysis, and machine-learning-driven optimisation of steganographic parameters. This section surveys these developments and positions the proposed method within the contemporary landscape.

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\subsection{Modern Advances in Reversible Data Hiding for General Images}

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\subsubsection{Prediction-Based High-Fidelity Embedding}

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Difference expansion \cite{11}, histogram shifting \cite{18}, and interpolation-based methods \cite{24} provide the classical foundation of RDH.

Their recent descendants improve the capacity-distortion trade-off through

adaptive prediction.

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Puteaux and Puech \cite{Puteaux2018MSB} proposed a reversible data hiding scheme based on most-significant-bit (MSB) prediction of pixel values, achieving near-lossless quality (PSNR above 51 dB) at payloads exceeding 0.5 bpp. The key insight is that the MSB of each pixel is highly predictable from its neighbourhood in natural images; embedding in the predictable MSBs rather than the less-predictable LSBs concentrates modifications where they are statistically expected and thus invisible. This prediction philosophy directly motivates the block-mapping approach: frequent patterns (PEAKs) are precisely those whose occurrence is predictable from the image's block statistics.

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Li et al. \cite{Li2019MultiHist} extend histogram shifting to multiple simultaneous histograms, identifying several peak-zero pairs rather than a single pair and embedding independently into each histogram. Their method achieves up to 0.48 bpp at PSNR above 43 dB on natural images, substantially outperforming single-histogram HS. The multi-histogram framework is structurally analogous to the proposed method's multi-PEAK strategy (\S\ref{subsec:multi_pattern}): by exploiting several frequent states simultaneously rather than a single dominant one, both methods increase capacity without proportionally increasing per-element distortion.

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\subsubsection{Surveys of the Contemporary RDH Landscape}

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Shi et al. \cite{Shi2019Survey} provide a comprehensive survey of RDH techniques from 2003 to 2019, documenting the evolution from single-layer DE and HS to multi-layer, content-adaptive, and deep-learning-assisted methods. Their taxonomy identifies three persistent bottlenecks that the proposed adaptive block-mapping directly targets: static threshold selection (addressed by the \$\mu_i + \alpha\sigma_i\$ mechanism), uniform-region use (evaluated in Section~\ref{subsec:rf-layer}), and adaptive capacity allocation across image regions (addressed by the multi-PEAK strategy).

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\subsection{Perceptual Quality Metrics for Binary Document Images}

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The evaluation of visual distortion in binary document images requires metrics that account for spatially non-uniform perceptual sensitivity. A pixel flip at the centre of a text stroke produces far greater visual annoyance than an identical flip in a background region, but PSNR assigns both the same penalty. The community-standard measure for binary document image quality is the Distance-Reciprocal Distortion (DRD) measure \cite{3535}, which weights pixel modifications by an inverse-distance neighbourhood function normalised over non-uniform \$8\times8\$ blocks. The threshold \$\text{DRD} \leq 1\$ for negligible distortion derives from the human perceptual study accompanying the original DRD publication \cite{3535}.

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Recent binary RDH papers have consistently adopted DRD alongside PSNR as the dual quality benchmark. Dong et al. \cite{Dong2020Adaptive} report DRD values below 0.5 for their adaptive overlapping scheme, establishing this as the competitive target. Zhang et al. \cite{Zhang2022Magnification} report \$\text{DRD} = 0.08\$ average at the cost of quadrupled image size, confirming the inverse relationship between resolution increase and visual distortion. Yin et al. \cite{Yin2020PPOCP} achieve \$\text{DRD} = 0.23\$ at 1157-bit average capacity on the same benchmark dataset, providing a reference point for capacity-distortion trade-offs.

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Feng, Lu, and Sun \cite{Feng2015Secure} established that the perceptual impact of binary image modification is governed by run-length statistics and connectivity. Their distortion model, which penalises modifications that disrupt run continuity or create isolated pixels, provides a principled alternative to pure Hamming-distance-based candidate selection. Although this paper predates 2018, its distortion framework was adopted and extended by subsequent works including \cite{Dong2020Adaptive, Zhang2022Magnification, Yin2020PPOCP}, making it an essential reference for contextualising candidate selection criteria.

\subsection{Pattern-Based RDH for Binary Images}

\subsubsection{The Block-Mapping Paradigm and Its Variants}

The block-mapping RDH paradigm partitions a binary image into non-overlapping \$3\times3\$ blocks, identifies frequent (PEAK) and absent (ZERO) patterns from the block histogram, and embeds data by mapping PEAK blocks to ZERO blocks within a controlled Hamming distance. Huynh and Nguyen \cite{Huynh2025BlockMapping} establish the current state of the art in this paradigm, achieving an average capacity of 2070 bits on eight standard \$512\times512\$ benchmark images using a fixed Hamming distance threshold. Their method exploits the combinatorial structure of the block pattern space more systematically than previous approaches but is constrained by the static threshold and the exclusion of uniform blocks.

Chhajed and Garg \cite{Chhajed2023BlockDiagonal} generalise the block representation to diagonal partition patterns, demonstrating that the choice of within-block structure creates a design space for capacity-distortion trade-offs. By restricting embedding to diagonally symmetric configurations, their method reduces the number of modified pixels per block while preserving reversibility. This work confirms that the \$3\times3\$ full-index encoding used by Huynh-Nguyen and the proposed method is not the only viable block representation.

\subsubsection{Overlapping and Sliding-Window Approaches}

An alternative to fixed non-overlapping blocks is the use of overlapping windows that scan the image to identify eligible patterns more densely.

Yin et al. \cite{Yin2020PPOCP} operate on overlapping \$3\times3\$ blocks and identify pattern pairs with opposite centre pixels (PPOCP), embedding one bit per qualifying block by potentially flipping the centre pixel. The method achieves 1157-bit average capacity with a PSNR-DRD profile that is more conservative than block mapping but provides finer control over which pixels are modified. PPOCP's eligibility criterion—requiring the simultaneous existence of both patterns in the image—focuses embedding on images with sufficiently rich pattern distributions.

Zhang et al. \cite{Zhang2022Magnification} take a fundamentally different approach by doubling the image resolution through \$2\times2\$ block expansion, embedding data in the expanded pixels while preserving the original pixel positions. Although achieving the highest PSNR (32.31 dB average) of all compared methods, the \$4\times\$ storage overhead renders this approach impractical for archival and transmission applications. A fair capacity comparison must normalise by image size; the raw 369-bit capacity at quadrupled storage represents a lower effective rate than its absolute figure suggests.

\subsection{Additional Binary-Image Methods}

The recent literature includes three binary RDH methods that are methodologically relevant to the proposed approach but were absent from the original comparative evaluation.

Dong et al. \cite{Dong2020Adaptive} provide the closest overlapping-pattern competitor: they adapt window placement to local texture, whereas the proposed

method adapts candidate assignment within a fixed block traversal. Their method is therefore included in the executable comparison rather than described again as a contextual-only baseline.

Ren et al. \cite{Ren2022Sliding} present RDH in binary images via a sliding window that scans horizontal runs, identifying eligible patterns along image lines rather than within fixed block boundaries. The sliding approach captures partial overlaps between frequently occurring short patterns, achieving higher capacity on text-heavy images with rich horizontal texture.

Yang \cite{Yang2023Large} targets large-capacity data hiding specifically in the black-and-white mixed regions of binary images, i.e., regions where both pixel values appear in close spatial proximity. Rather than treating all non-uniform blocks identically, Yang stratifies blocks by local black-white mixing ratio and assigns higher embedding priority to those with the most balanced mixture. This density-based prioritisation is complementary to the proposed method's distance-based adaptive threshold.

Dong is included in the executable common protocol of Section~\ref{sec:experimental}. Ren, Yang, Zhang, and Chhajed-Garg remain contextual references because no validated author implementation was available and their publications do not fully specify an independently serializable synchronization stream under the net-capacity definition used here. Zhang additionally changes image dimensions, so its storage rate is not commensurate with the fixed-size protocol. Table~\ref{tab:modern_comparison} reports their published operating points without treating them as wire-level baselines.

\subsection{Privacy-Preserving and Encrypted Binary Image RDH}

A growing body of recent work extends binary RDH to scenarios where the data hider does not have access to the plaintext image—a deployment model directly relevant to cloud storage and distributed medical imaging systems.

Ren, Lu, and Chen \cite{Ren2019Encrypted} propose RDH in encrypted binary images by pixel prediction. The content owner encrypts the binary image with a stream cipher; the data hider then identifies positions where a local neighbourhood predictor is highly confident and flips a subset of encrypted pixels to embed the message. The receiver, holding the encryption key, decrypts and extracts embedded bits from the pre-agreed positions. This work establishes the feasibility of binary RDH in the encrypted domain.

Ma et al. \cite{Ma2021RDHEncrypted} extend encrypted image RDH to multi-key scenarios where different parties hold partial decryption keys, enabling hierarchical access to embedded data. Their construction is directly relevant to medical and legal document applications: a hospital image archive could allow radiologists to embed annotations via RDH without ever decrypting the full patient record.

Recent encrypted-domain research also includes adaptive block coding for grayscale carriers \cite{Xiao2024AdaptiveBlock}. It is relevant to side-information compression, but its multi-bit grayscale cover model and vacating-room-after-encryption protocol are not commensurate with reversible mapping in binary cover images. Within the direct binary block-mapping scope, Huynh-Nguyen \cite{Huynh2025BlockMapping} is the recent 2025 continuation used in the present positioning.

\subsection{Deep Learning Steganalysis of Binary Document Images}

For applications in sensitive domains (medicine, law, administration), resistance to detection is a primary security concern. The steganalysis community has shifted decisively since 2018 from hand-crafted feature sets to convolutional neural networks trained end-to-end on raw pixel values.

Boroumand, Chen, and Fridrich \cite{Boroumand2019SRNet} introduced SRNet, a deep residual network for image steganalysis that operates directly on image residuals rather than features engineered for a specific embedding algorithm. Its success motivates learned binary-image steganalysis, although a validated binary adaptation requires a dedicated architecture and substantially more cover-stego pairs than are available in the present protocol.

Yedroudj, Comby, and Chaumont \cite{Yedroudj2018Net} propose Yedroudj-Net, which augments a convolutional backbone with a rich-model preprocessing layer, combining the interpretability of hand-crafted features with the power of learned representation. Yedroudj-Net is particularly effective at detecting block-based embedding because the preprocessing layer explicitly captures block-boundary artefacts—the statistical signature introduced by non-overlapping 3×3 block mapping.

Zhang et al. \cite{Zhang2019SRNet} extend depth-wise separable convolution architecture to steganalysis, achieving comparable detection accuracy to SRNet at substantially lower computational cost. Their architecture's efficiency makes it practical to evaluate detection rates across many embedding parameters (α values, payload sizes).

Block-pattern-mapping RDH creates a predictable statistical signature in the block histogram: the stego image has reduced counts at PEAK patterns and increased counts at ZERO patterns relative to the cover image. The companion adaptive steganography work from the same research group \cite{Ekodeck2026Combi} demonstrates that the KL divergence between cover and stego pattern distributions is the primary statistical footprint of block-mapping steganography and that reducing it substantially decreases detectability.

\subsection{Adaptive Parameter Selection and Optimisation}

The adaptive-threshold model $\text{HT}_i=\mu_i+\alpha\sigma_i$ is used to characterize the unconstrained capacity-distortion landscape. The deployed system then adopts the distance-one subset of this design space because it gives a deterministic one-bit mapping, bounded pixel distortion, and a compact auxiliary codec.

Frazier \cite{Frazier2018Bayesian} provides a comprehensive tutorial on Bayesian optimization with Gaussian-process priors. The normalized objective defined in \S\ref{subsec:param_optim} is a non-convex and evaluation-intensive function of the type targeted by this framework.

Sedighi, Cogranne, and Fridrich \cite{Sedighi2018Content} establish that optimal steganographic embedding assigns each cover element an individual distortion cost based on local image statistics, rather than using a uniform threshold. Their content-adaptive framework—based on the multivariate Gaussian model of local image windows—is directly applicable to binary block-mapping: each PEAK pattern P_i should receive a per-pattern distortion function calibrated to its specific neighbourhood statistics.

Jia et al. \cite{Jia2019Guided} propose a guided image steganography framework that simultaneously optimises multiple competing objectives (capacity, distortion, and security) through a multi-objective evolutionary algorithm. Applied to binary block-mapping, this framework would avoid the single scalar λ weighting by instead maintaining a Pareto front of $(C_{\text{total}}, \text{DRD})$ trade-off points.

\subsection{Machine Learning Integration in Steganography and Reversible Watermarking}

A substantial body of recent work now applies machine learning directly to steganographic embedding, steganalysis, and watermarking.

Zhu et al. \cite{Zhu2018HiDDeN} introduced HiDDeN, the first end-to-end trained encoder-decoder steganographic system where an encoder network embeds a fixed-length binary message into a cover image, a decoder recovers the message, and a discriminator simultaneously trains to detect the stego image. The adversarial training naturally balances capacity, imperceptibility, and security without requiring an explicit distortion metric.

Jing et al. \cite{Jing2021HiNet} propose HiNet, an invertible neural network architecture that achieves lossless information hiding by design: the network is architecturally reversible, guaranteeing perfect extraction without a separate reversibility mechanism. For binary RDH, an analogous invertible architecture operating on block-pattern features could replace explicit bijective mapping tables with a learned, parameter-free reversible transform.

Lu et al. \cite{Lu2021LargeCapacity} substantially increase the capacity of invertible image hiding by replacing scalar message embedding with a full second image hidden within the cover. Their architecture hides an entire image within another image of identical resolution, achieving image-in-image steganography at PSNR above 33 dB. While targeting natural greyscale images, the architectural principles transfer to binary images.

Li, Feng, and Zhang \cite{Li2021LearnedRDH} demonstrate that a learned distortion predictor for RDH—trained on pairs of (cover block, modified block, human annoyance rating)—substantially reduces visual distortion relative to raw distance ordering at the same embedding capacity. This approach is directly applicable to improving candidate ZERO pattern selection in block-mapping methods.

\subsection{Comparative Summary and Research Gap Analysis}

Table~\ref{tab:modern_comparison} consolidates the literature across the
evaluation dimensions that determine practical reversible payload. Existing
binary-image studies generally report gross embedding capacity without the
serialized synchronization cost, so their published values do not establish
how many application bits remain after exact extraction metadata. They also
rarely combine systematic steganalysis with capacity-distortion reporting, and no identified study evaluates several representative methods using the
same covers, messages, payload targets, distortion implementation, and wire-level accounting. The resulting gap is therefore both methodological and
experimental: binary RDH lacks a common net-capacity protocol that preserves
exact reversibility while measuring perceptual, computational, and detection
trade-offs. The proposed framework and evaluation protocol are designed to
fill this gap.

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Yin et al. 2020 \cite{Yin2020PPOCP} & 1157 & Not reported & 27.17 & 0.23 & No & Excluded & No \\

Dong et al. 2020 \cite{Dong2020Adaptive} & $\sim$1500 & Not reported & $\sim$25 & $<$0.5 & Yes & Excluded & No \\

Ren et al. 2022 \cite{Ren2022Sliding} & $\sim$2000 & Not reported & $\sim$24 & $<$1 & No & Excluded & No \\

Zhang et al. 2022 \cite{Zhang2022Magnification} & 369 & Not reported & 32.31 & 0.08 & No & Excluded & No \\

Chhajed \& Garg 2023 \cite{Chhajed2023BlockDiagonal} & $<$2000 & Not reported & $\sim$25 & $<$1 & No & Excluded & No \\

Yang 2023 \cite{Yang2023Large} & $>$2000 & Not reported & $\sim$23 & $<$1 & Partial & Excluded & No \\

Huynh \& Nguyen 2025 \cite{Huynh2025BlockMapping} & 2070 & Not reported & 24.13 & 0.55 & No & Excluded & No \\

\textbf{Proposed (this work)} & \textbf{2874 gross} & \textbf{2245 net} & \textbf{per-image} & \textbf{0.76} & \textbf{Yes} & \textbf{Evaluated} & \textbf{Yes} \\

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\caption*{EC: capacity reported by the cited study on its stated 512×512 protocol. Values are contextual rather than directly comparable because datasets, payload definitions, and side-information accounting differ. “Steg. eval.”: steganalysis resistance evaluated.}

Deleted block 24

Section~\ref{sec:experimental}. It extends the block-mapping framework of
Huynh and Nguyen~\cite{Huynh2025BlockMapping} with learned candidate ordering and explicit
optimization of payload after serialized metadata. The unconstrained adaptive
threshold is retained as an ablation that characterizes the broader capacity-distortion landscape; the deployed codec uses globally disjoint Hamming-distance-one substitutions. Consequently, each active block carries
one bit and changes at most one pixel.

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The system:

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\item selects the frequency-ordered table prefix that maximizes net capacity; and

Deleted block 27

\[

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\]

The product, rather than the sum, gives the block count:

\[

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\]

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\[

d_k = \sum_{i=0}^{8} b_i 2^i,

\]

where $b_i \in \{0,1\}$ follows a fixed row-major ordering shared by both embedding and extraction.

Rows below $3\lfloor M/3\rfloor$ and columns beyond
$3\lfloor N/3\rfloor$ are copied unchanged and never participate in capacity
or extraction.

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\item \textbf{Residual patterns}: patterns not involved in embedding.

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net-capacity rule defined below, not by an unspecified histogram tail.

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The candidate set is defined as:

\[

\mathcal{Z}_i = \{ Z_j \in \mathcal{Z} \mid D_{i,j} \le \mu_i + \alpha \sigma_i \},

\]

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This formulation relies solely on empirical statistics and does not assume any specific distance distribution.

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Each block is modified at most once, and no inter-block dependencies are introduced.

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extraction; for the deployed one-bit codec, no padding ambiguity arises.

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\node[process, left=of inject] (net) {Sort PEAKs by frequency\\select prefix maximizing\\$C_{\rm gross}-|A_{\rm wire}|_2$};

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followed by explicit net-capacity optimization and enumerative serialization;

no cryptographic operation is assumed.}
Deleted block 49
$\sum_{P_i\in\mathcal T^\star}H(P_i)-|A_{\rm wire}(\mathcal T^\star)|_2$
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\draw[arrow] (done) – node[above]{No} (lookup);
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\[
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\]
Deleted block 53
\State reject the malformed stego/auxiliary pair
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\[
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\]
Deleted block 56
\[
Deleted block 57
\]
Expected Improvement selects the next value of α. The equal weighting
Deleted block 58
it is not used to select or train the deployed distance-one system. A λ sweep is consequently not used to support any reported deployment
claim: changing this diagnostic scalarization would alter only the exploratory
ordering, whereas all final tables use the independently specified
distance-one policy. Optimizing α alone does not guarantee
Deleted block 59
integrated system. The exploratory search uses
Deleted block 60
151-point grid, $\lambda=0.5$, exploration 0.01, and a Matérn-$5/2$ kernel
Deleted block 61
9×9 channels: the centred PEAK, the centred candidate, and the average
Deleted block 62
cost is assigned to symbol one, while symbol zero preserves the PEAK.
Deleted block 63
embedding block; a 7×7 patch cannot provide that support for corner
flips. Larger patches add context that is not used by the local training target.
Deleted block 64
and seed 20260607. Complete source images, rather than candidate samples, define
the train/validation split.
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safety label uses a local reciprocal-distance cost no greater than 0.5.
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incurs coordinate overhead. They were not optimized on the reported test
outcomes. This layer is an evaluated ablation and is not included in the final
payload figures because its measured incremental net contribution is negative.
Deleted block 67
Useful capacity subtracts the exact serialized side information:
\[
C_{\mathrm{net}}=C_{\mathrm{gross}}-|A_{\mathrm{wire}}|_2.
\]
Deleted block 68
positions are then represented as base-9 digits. This approaches the
information content of the mapping set more closely than independent fixed
width pairs. Mapping pairs are ordered by occurrence count, and only the prefix
maximizing net capacity is retained. If no positive subset exists, the image is
left unchanged.
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\[
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Deleted block 73
The wire stream contains the magic identifier, format version, 32-bit payload
length q, 16-bit table count t, and the shortest fixed byte string capable
of representing the state space $\binom{512}{t}9^t$. Image dimensions are
provided by the stego container and are checked by the extraction API. The
stream is compressed but not encrypted; confidentiality and authentication,
when required by an application, are orthogonal transport-layer services.
Deleted block 74
Compared with~\cite{Huynh2025BlockMapping}, the proposed scheme introduces adaptive candidate selection,
pattern-aware capacity allocation, and multi-pattern embedding,
resulting in improved embedding efficiency while preserving reversibility.
On the eight document benchmarks, the proposed implementation provides 2874
gross bits and 2245 net bits on average after 629 auxiliary bits are charged.
Huynh-Nguyen report 2070 gross bits under their own protocol; this published
value is a contextual operating point rather than a direct net-capacity
comparison because their serialized auxiliary cost is not reported.
Deleted block 75
This subsection provides a formal analysis of the computational complexity and
theoretical guarantees of the proposed adaptive block-mapping embedding and
optimized extraction algorithms. In particular, we derive tight asymptotic
bounds for both time and space complexity, and we establish key properties such
as deterministic reversibility and absence of inter-block interference. These
results ensure that the proposed scheme is not only efficient but also
theoretically sound and suitable for high-integrity reversible data hiding
applications.
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\begin{proposition}[Independent Blocks]
Embedding operations on distinct blocks are independent.
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Each block is processed independently using a pattern-specific mapping table.
No embedding operation modifies neighboring blocks or depends on the state of
other blocks.
Therefore, embedding decisions are local and do not introduce inter-block interference.
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test images. The profile split supplies only the image-level statistics needed
by the conservative Dong implementation. Test images are binarized at
Deleted block 87
without BOSSbase retraining, preventing test-set adaptation.
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128, 256, and 512 bits. Every available run is accepted only after exact
message extraction and bitwise cover restoration. Huynh-Nguyen uses the
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publications leave the independent wire representation open. These
implementations are not presented as official author code. Accordingly,
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study, while DRD and availability provide the less
serialization-dependent comparison.
Deleted block 91
Gross payload is the message length, auxiliary bits are the exact serialized
wire length, and useful payload is
$C_{\mathrm{net}}=C_{\mathrm{gross}}-C_{\mathrm{aux}}$. Distortion is measured
with changed pixels, PSNR, and DRD \cite{3535}. Availability is the fraction of
test images that carry the target and complete the exact round trip.
Deleted block 92
image-level resamples and seed 20260608. Runtime includes embedding,
Deleted block 93
exactly. The previously defined two-image split gives MAE 0.081 and correlation
0.950 and is retained only as a directly reproducible detailed case:
CNN ranking reduces DRD from 0.825 to 0.795 on table1-3 and from 0.935 to 0.899
on french-4. Thus, regression accuracy and operational DRD impact are evaluated
separately, and the latter is not inferred from two selected examples.
Fold-level regression and placement results are provided with the
reproducibility artefacts.
Deleted block 94
recall 0.970, and flip-position accuracy 0.971. The layer adds 179.8 gross bits
per document image on average, while its current coordinate representation
uses 2667.0 auxiliary bits. This identifies coordinate coding as a concrete
optimization opportunity while retaining the classifier as a validated component.
Deleted block 95
On the 100 BOSSbase test images, net-aware table selection gives 893.2 gross,
377.1 auxiliary, and 516.1 net bits on average. Median net capacity is 362
bits, 78 images are net positive, mean DRD is 0.402, and every evaluated
message and cover is recovered exactly. The enumerative codec raises external
net capacity by 32.1\% relative to the preceding dense-or-sparse wire format.
Deleted block 96
all-method matched subsets. ABM moves from metadata-dominated operation at 64
bits to a median-positive point at 128 bits and a clearly positive operating
region at 256 and 512 bits. PPOCP defines the lowest-DRD trajectory, while ABM
defines the highest-net trajectory.
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\resizebox{\textwidth}{!}{
\begin{tabular}{@{}rrcccccccc@{}}
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\end{tabular}}
Deleted block 99
64 & Proposed ABM & $-48.5\;[-49.2,-48.0]$ & 0.022\\
& Dong / Huynh-Nguyen / PPOCP & $-226.3\;[-229.6,-223.1]$ /
$-1644.7\;[-1855.7,-1450.9]$ / $-1028.0\;[-1063.1,-989.2]$ &
Deleted block 100
& Dong / Huynh-Nguyen / PPOCP & $-178.3\;[-188.2,-169.7]$ /
$-1582.8\;[-1794.4,-1386.3]$ / $-1707.8\;[-1767.1,-1645.0]$ &
Deleted block 101
256 & Proposed ABM & $113.9\;[108.5,119.0]$ & 0.087\\
& Dong / Huynh-Nguyen / PPOCP & $-83.8\;[-106.1,-66.1]$ /
$-1456.3\;[-1662.6,-1264.0]$ / $-2954.1\;[-3064.0,-2831.5]$ &
Deleted block 102
512 & Proposed ABM & $309.0\;[289.7,326.1]$ & 0.176\\
& Dong / Huynh-Nguyen / PPOCP & $95.8\;[55.8,130.6]$ /
$-1190.6\;[-1396.5,-994.8]$ / $-4821.7\;[-5028.9,-4612.9]$ &
Deleted block 103
payload-specific matched analysis. It also removes changing-cohort composition
as an explanation for the positive ABM net payload from 128 bits onward.
Deleted block 104
(left) and median DRD (right).}
Deleted block 105
the net-bits-DRD operating trajectories. PPOCP carries every target on all
Deleted block 106
The right panel is descriptive rather than a scalar ranking because the
methods span different DRD ranges; Table~\ref{tab:fixed-cohort} provides the
Deleted block 107
efficiency. Numbers on the right panel denote target payloads in bits.}
Deleted block 108
3016 bits. DRD tests place ABM below Dong and above the perceptually focused
PPOCP and Huynh-Nguyen trajectories, yielding a two-objective interpretation.
Deleted block 109
Suitability therefore acts as an applicability indicator rather than a
post-hoc explanation of net performance.
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34.2, and 0.024. Unavailability is therefore concentrated in near-empty or
nearly uniform thresholded photographs with insufficient repeated
non-uniform patterns, rather than in failed extraction or restoration.
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most resistant trajectory from 128 to 512 bits. ABM's AUC increases with useful
payload, making statistical footprint the principal optimization signal for
the next design iteration. The Random Forest gives ABM AUC values 0.623, 0.691,
Deleted block 112
comparisons. In useful capacity, ABM establishes the highest observed

trajectory and becomes mean-net-positive between 128 and 256 bits. In
Deleted block 113
least-detectable trajectory. This conclusion rests on exact round trips,
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not presented as a universal ranking of the original publications.
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The measured AUC curves provide a direct route to strengthen the method. The
current CNN predicts reciprocal-distance cost, whereas the steganalysis
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\[
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The present measurements define an operational distinction. At 256 and 512 bits, logistic AUC values of 0.900 and 0.957 make the current ABM configuration unsuitable when covert undetectability is the primary requirement, despite its positive net payload and exact reversibility. Its
validated contribution at these loads is capacity-oriented reversible annotation, not secure covert communication. At 64 bits, AUC 0.706 provides a
less detectable but metadata-negative operating point. PPOCP remains the
Deleted block 119
The RF uniform-block experiment supplies a second concrete direction:
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binarization is a transfer test rather than a substitute for a large scanned
document corpus. The 512-bit all-method cohort contains 46 images, so both
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visual DRD remains moderate. These measured boundaries delimit the validated
Deleted block 122
The reversible channel assumes lossless storage and transmission; geometric
or lossy transformations require an external synchronization or error-control
layer.
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This work develops and evaluates a net-aware, machine-learning-assisted block-mapping framework for exactly reversible data hiding in binary images.
The CNN ranker improves candidate placement on held-out images, the enumerative codec reduces auxiliary cost, and the complete system carries 2245
net bits on average on eight document images and 516.1 net bits on 100 BOSSbase images. Every reported available experiment restores the message and
cover exactly.
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6.7×10^{-13}. Availability, binarization, runtime, memory, and two-classifier
steganalysis turn this result into a reproducible practical profile rather than
a single capacity number.
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evidence identifies ABM as the highest observed net-payload operating point,
while PPOCP retains the strongest measured distortion and steganalysis
profile. The result therefore adds a concrete net-capacity benchmark and an
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also identify local-distribution preservation as the principal optimization
Deleted block 127
This work was supported by the authors' institutions. No external financial
support was received.
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personal relationships that could have appeared to influence the work
reported in this paper. The companion publication cited in the related work
shares authors with this study and is identified explicitly; it is not used
as an experimental baseline or as evidence for the numerical claims reported
here.
Deleted block 129
FUNSD update – replaced abstract material
document images, this activation
raises mean net payload from 2139 to 2245 bits relative to an otherwise
identical wire-charged gross-selection variant and preserves exact round-trip
reversibility. In a common executable protocol on 100 disjoint thresholded
BOSSbase images, the proposed adaptive block-mapping method (ABM), Dong,
Huynh-Nguyen, and the opposite-center-pair method (PPOCP) are compared with
identical messages, payload targets, distance-reciprocal distortion (DRD)
computation, auxiliary-bit accounting, and reversibility checks. The proposed
method achieves the highest useful
capacity among the evaluated implementations, while PPOCP retains the
lowest-distortion and most favorable detectability profile. Logistic detector AUC reaches
0.900 and 0.957 at 256 and 512 bits, placing the validated operating region in
capacity-oriented reversible annotation over a lossless channel with
detectability-aware payload selection.
FUNSD update – replaced roadmap material
describes the common protocol and reports document-image and BOSSbase results.
Section \ref{sec:discussion} synthesizes the practical operating region, and
Section \ref{sec:conclusion} concludes the study.
FUNSD update – replaced dataset/protocol material
The evaluation has two complementary scales. The eight standard
512-by-512 binary document images used by Huynh-Nguyen support direct
validation against the block-mapping literature, while BOSSbase supplies the
disjoint profile/test transfer protocol. Test images are binarized at threshold
128 and the CNN ranker is transferred directly to BOSSbase.
The four-method protocol embeds identical pseudorandom messages at targets 64,
128, 256, and 512 bits.
FUNSD update – replaced BOSSbase interpretation
all-method matched subsets. ABM moves from metadata-dominated operation at
64 bits to a near break-even 128-bit point and reaches positive mean net
payload at 256 and 512 bits.
FUNSD update – replaced experimental-scope material
The eight document images provide continuity with the binary block-mapping
literature, and the disjoint BOSSbase protocol adds 100 external test images,
four payloads, paired inference, resources, and steganalysis. BOSSbase
binarization should be read as a transfer test, with a large scanned document
corpus remaining a natural next benchmark. The 512-bit all-method cohort

contains 46 images, so both availability and fixed-cohort results are reported.
The reversible channel assumes lossless storage and transmission; geometric,
noisy, or lossy transformations require an external synchronization or error-control layer.
FUNSD update - replaced conclusion material
The complete system delivers 2245 net bits on average on eight document images
and 292.5 net bits on 100 BOSSbase images. Every reported available experiment
restores the message and cover exactly. Under the common protocol, ABM obtains
mean net payloads of -57, -7, 93, and 309 bits at targets 64, 128, 256, and
512 bits on matched images. The evidence adds a concrete net-capacity benchmark
and an exactly reversible reference implementation, with local-distribution
preservation as the main optimization target.
FUNSD update - replaced data-availability material
BOSSbase is publicly available from its original distributor under its
applicable terms. The eight document images used for block-mapping validation,
derived aggregate results, and per-image evaluation artefacts accompany the
reproducibility package.

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